

APPROVED METHOD:
OPTICAL AND ELECTRICAL
MEASUREMENTS
OF LED SOURCES
AN AMERICAN NATIONAL STANDARD

ANSI/IES LM-85-23

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OF LED SOURCES
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has been approved by IES.
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should be directed to IES.

**Prepared by
The IES Testing Procedures Committee**



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1.0 Introduction and Scope

1.1 Introduction

This document provides guidance for the measurement of light emitting diode (LED) sources such as LED packages and LED arrays. The output of an LED source is strongly dependent on its operating conditions—in particular, the junction temperature (T_j). Minor changes in T_j can significantly affect the output of an LED. Therefore, accurately setting and measuring T_j is critical for photometric and colorimetric measurements. LED manufacturers typically provide performance characteristics of LED sources at T_j of 25 °C and/or 85 °C.

LEDs' photometric and radiometric characteristics for lighting applications are typically total luminous flux (lumens), total radiant flux (watts), total photon flux (moles per second, mol/s) for horticultural lighting, color characteristics for white LEDs, and spectral characteristics for single-color LEDs. The color characteristics include chromaticity coordinates, correlated color temperature (CCT), distance from the Planckian locus (D_{uv}), color rendering index (CRI), and other color quantities such as color fidelity metric, R_i , described in ANSI/IES TM-30-20.¹ The spectral characteristics include dominant wavelength, centroid wavelength, and peak wavelengths. For this document, the determination of these values is referred to as *optical measurements*.

The electrical characteristics for lighting applications are typically LED forward current, I_F , forward voltage, V_F , and input power. For this document, the determination of these values is referred to as *electrical measurements*.

Four different measurement methods—single pulse, flash pulse, continuous pulse, and DC—are described in this document.

1.2 Scope

This Approved Method describes the procedures for optical and electrical measurements of LED sources. The methods in this document cover white and single-color LED sources and may also be applied to laser diode sources; however, laser diodes may have special drive circuitry requirements, which should be obtained from the manufacturer.

This document does not cover ultraviolet LED sources (covered in ANSI/IES LM-92-22), alternating-current (AC) LED sources (covered ANSI/IES LM-88-18(R23)), or LED lighting products (covered in ANSI/IES LM-79-19). In addition, it does not cover relative measurement methods.

2.0 Normative References

2.1 ANSI/IES LS-1-22

Illuminating Engineering Society. Lighting Science: Nomenclature and Definitions for Illuminating Engineering. New York: IES; 2022. Online: www.ies.org/standards/definitions/. (Accessed 2023 Oct 5).

2.2 ANSI/IES LM-58-20

Illuminating Engineering Society. Approved Method: Spectroradiometric Measurement Methods for Light Sources. New York: IES; 2020.

2.3 ANSI/IES LM-75-19

Illuminating Engineering Society. Approved Method: Guide to Goniometer Measurements and Types, and Photometric Coordinate Systems. New York: IES; 2019.

2.4 ANSI/IES LM-78-20

Illuminating Engineering Society. Approved Method: Total Luminous Flux Measurement of Lamps Using an Integrating Sphere Photometer. New York: IES; 2020.

2.5 ISO/IEC Guide 98-3:2008

International Electrotechnical Commission, Uncertainty of Measurement – Part 3: Guide to the expression of uncertainty in measurement (GUM:1995). Geneva: IEC; 2008.

2.6 JCGM 104:2009

International Bureau of Weights and Measures (BIPM); Joint Committee for Guides in Metrology (JCGM). Evaluation of measurement data — An introduction to the “Guide to the expression of uncertainty in measurement” and related documents. Sèvres, France: BIPM; 2009.

3.0 Definitions

ANSI/IES LS-1-22 (see **Section 2.1**) definitions apply unless otherwise defined below.

3.1 device under test (DUT)

For this document: An LED source, a remote-phosphor LED source, or laser diode light source under test.

3.2 junction temperature (T_J)

Temperature of a DUT p-n junction; in the case of multiple junction DUTs, the T_J is considered as a single-junction equivalent that represents the effective (or aggregated) temperature.

3.3 LED source

An LED package, array, or module (including filament LEDs) that is operated via an auxiliary driver.

3.4 remote-phosphor LED or remote-phosphor laser diode source

An LED source containing a component with one or more phosphor materials that is mechanically separated from its pump LED die but is contained within the LED source. There is a gas or other suitable material between the LED die and the phosphor component, which leads to a high thermal resistance between the pump LED die and the phosphor component.

3.5 settling time

The time needed for the current signal to reach a stable level after a fast transition; it is a function of the power supply, DUT, and cabling.

3.6 temperature coefficient of forward voltage, S_{VF}

Sensitivity factor of the forward voltage to junction temperature, S_{VF} (V/K). (Note: S_{VF} is a negative value.)

3.7 temperature-controlled mount (TCM)

A device that incorporates one or more cooling and/or heating components to maintain its temperature at a set value. A TCM is used to control the temperature of an LED mounted on it for testing purposes. A thermoelectric cooler (TEC) is often used for a TCM. A TCM is sometimes called a *temperature-controlled heat sink*, though it may provide heat to a DUT.

3.8 uncertainty

In this document, all uses of the word *uncertainty* refer to expanded uncertainty with a 95% confidence interval, normally with a coverage factor of $k = 2$, as prescribed by standards. If the manufacturer's specification does not specify uncertainty as described here, then the manufacturer should be contacted for proper conversion.

4.0 Physical and Environmental Conditions

4.1 Ambient Temperature, T_A

Ambient temperature, T_A , should be $25\text{ }^{\circ}\text{C} \pm 2\text{ }^{\circ}\text{C}$.

T_A shall be measured at a point not more than 0.5 m from the DUT and at the same height as the DUT. The temperature sensor shall be shielded from direct radiation from the DUT.

4.2 Humidity

Relative humidity values greater than approximately 65% can lead to corrosion effects in some instruments, and values below approximately 10% can lead to electrostatic effects. Therefore, laboratory humidity should be monitored and maintained between 10% and 65%.

4.3 Airflow

The incidence of air movements on the surface of the DUT may alter the T_J . Airflow around the DUT should be such that normal convective airflow is not disturbed.

4.4 Vibration

DUTs should not be subjected to excessive vibration or shock during stabilization, transportation, mounting, or testing.

5.0 Electrical Measurement Instrumentation

5.1 Instrument Types

A pulsed current source (PCS), and/or a pulsed source

measure unit (P-SMU), data acquisition units (DAQs), current shunt or current probe, digital multimeter (DMM), oscilloscope, and TCM are typically required.

5.1.1 PCS. The PCS provides the direct current (DC) or pulsed current that powers the DUT during measurements. The PCS shall be able to source constant current at the required amplitude, with programmable pulse width capability in the range of 10 μ s to 50 μ s (10 ms to 50 ms for the single-pulse methods), and programmable pulse period capability that supports duty cycles in the range of 1% to 20%. The PCS compliance limit shall be greater than the DUT's maximum V_F plus any cable voltage drops. To produce a fast-rising pulse, some PCSs employ an amplifier to apply a voltage that is much higher than a DUT's V_F (e.g., 20 V to 30 V higher). This voltage is used to overcome cabling inductance. The PCS power limitation shall be greater than the maximum required DUT electrical power during the measurement.

The total PCS timing error, $t_{\text{total-err}}$, shall not exceed the width of the short pulse applied as part of the continuous-pulse method. For a typical PCS this is:

$$t_{\text{total-err}} = t_{\text{pos-settling}} + t_{\text{pw}} + t_{\text{jitter}} + t_{\text{neg-settling}} < 10 \mu\text{s} , \quad (5-1)$$

where:

$t_{\text{pos-settling}}$ = the positive transition settling time

t_{pw} = the pulse width error

t_{jitter} = the pulse width jitter error

$t_{\text{neg-settling}}$ = the negative transition settling time

In addition, PCS pulse width timing jitter shall not exceed 1% of the pulse width. The PCS period jitter shall not exceed 1% of the period.

PCS DC current accuracy shall be assessed to ensure the current error contribution to the measurement uncertainty is within acceptable limits. In any case, the current uncertainty (after settling) shall be better than 0.5%.

5.1.2 P-SMU. A P-SMU is a source measurement unit with capabilities of sourcing and measuring pulses. P-SMUs typically have a hardware-based pulse generation

capability that supports microsecond pulsing, and they include deterministic hardware-based timing systems that can meet the error and jitter requirements. General-purpose SMUs without hardware-based pulse generation are not suitable instruments to measure DUTs under a pulse operation.

5.1.3 DAQ. DAQs are often used for V_F and I-V sweep measurements. These measurements require the DUT's V_F signal to be averaged over a defined portion of the pulse. DAQs normally feature fast sampling and excellent trigger resolution; however, they may lack flexibility in measurement aperture programming. If this is the case, samples should be collected and averaged post-measurement. Care should be taken when selecting a DAQ to ensure that it has adequate common mode voltage capability. The DAQ common mode specification should be reviewed prior to selection to ensure that it will not be exceeded. DAQs used for V_F and I-V sweep measurements shall have an analog-to-digital converter (ADC) sampling resolution and overall accuracy suitable for making voltage measurements with the required uncertainty. In any case, the uncertainty of the voltage measurement shall be less than 0.2%, and the bandwidth should be higher than 400 kHz. For lowest noise, it should have a floating, differential input. (Note: A DAQ may be used to visualize the overall current waveform, but if it has a limited bandwidth and sample rate, it shall not be used to measure or correct the waveform.)

Some P-SMUs include a built-in DAQ. Using such a P-SMU simplifies pulse generation and measurement coordination.

Some DMMs also have a built-in DAQ and have programmable trigger and aperture functions. If such a DMM is used, it shall be linked to the PCS with a hardware trigger.

5.1.4 Current Shunt or Current Probe. These devices sense the current through the DUT, and they produce a voltage signal that is measured with a DMM or high-speed DAQ. Current shunts and probes should be attached near the DUT to allow the measurement to account for capacitive cable losses.

Suitable current probes are high-accuracy Hall-effect types. To minimize errors, the probe should be zeroed prior to a measurement, and the probe's amplitude error should be characterized and corrected. Probe bandwidth shall be 8 MHz or greater.

Suitable current shunts are built into many DMM instruments. External shunts may also be used. These are especially useful for currents above 10 A. Cable inductance to and from the shunt should be minimized by keeping wires as short as possible. If the shunt is connected to an instrument (DMM, DAQ, or oscilloscope) with an earth ground connection, some DC or AC current may flow back to the PCS through the ground connection, degrading the pulse accuracy. To avoid this, shunts should only be used with isolated instruments, or the shunt's output signal should be conditioned first with a differential amplifier prior to connecting it to the grounded instrument.

5.1.5 DMM. A DMM may be used for measuring average I_F and/or V_F using a hardware triggering and proper measurement aperture when a single pulse method is used. If the mean-current differential continuous-pulse method is used, then a 6½-digit DMM or better with a multi-slope integrator is required. These instruments measure the average of the pulsed current signal. This average is then used to correct the measurements.

Commercial benchtop DMMs typically have the required multi-slope integrator. This circuit integrates the incoming voltage signal to determine its average amplitude. Because the integrator analog circuitry responds to both high-frequency and low-frequency signal components, and because it has excellent accuracy, it is a good choice for measuring the average pulse current signal. DMMs also frequently include suitable current shunts.

High-speed DAQ instruments may also be used to measure the average of the current signal. If a DAQ is used, its sampling rate shall be greater than 10 mega samples per second, and its bandwidth shall be greater than 10 MHz.

5.1.6 Oscilloscope While the methods in this document do not require an oscilloscope for measurement,

it is required for visualizing the current waveform. Visualization allows the current waveform to be verified, and it is useful when adjusting the load-tuning settings available in some PCS. Current waveform visualization may also be achieved using a DAQ. If a DAQ is used, an external shunt or current probe should be used to minimize errors due to cable capacitance.

5.1.7 TCM. The measurement methods require the DUT to be maintained at a specific temperature. This is normally achieved by placing the DUT on a TCM. TCMs shall have a temperature resolution of 0.1 °C and a stability of 0.1 °C.

Commercial thermoelectric cooler (TEC)-based mounts are good choices for this function. These mounts may be equipped with mounting attachments that make DUT installation and removal easy. TEC-based mounts should be sized to handle the average power applied to the DUT.

5.2 Instrument Uncertainty

The uncertainties of the DMM for measurement of DC current (I_F) and voltage (V_F) shall be less than 0.1%. The uncertainties of the pulse current meter and pulse voltage meter, or SMU, varies depending on the pulse width, and in any case, shall be less than 1%.

6.0 Optical Measurement Instruments and Optical Measurements

6.1 Optical Measurement Instruments

The instruments described here are applicable to both the pulse-mode method and the DC-mode method. If the pulse-mode method is used, the photometer or the spectroradiometer shall have capability to measure optical pulses.

One or more of the following instruments are used for the measurement of photometric, colorimetric, and radiometric quantities:

- Sphere-spectroradiometers (for total luminous flux, total radiant flux, photon flux, and color quantities)

- Sphere-photometers (for total luminous flux)
- Goniospectroradiometers (for luminous intensity distribution, total luminous flux, total radiant flux, color quantities, and angular chromaticity uniformity)
- Goniophotometers (for luminous intensity distribution and total luminous flux)

A sphere-spectroradiometer is an integrating sphere (or an integrating hemisphere) equipped with a spectroradiometer as the detector. A sphere-photometer is an integrating sphere (or an integrating hemisphere) with a photometer head. A goniospectroradiometer is a goniometer equipped with a spectroradiometer. A goniophotometer is a goniometer equipped with a photometer head.

Array spectroradiometer-based sphere-spectroradiometers are recommended. They can acquire all optical quantities by a single measurement and do not have the spectral mismatch error. Thus, they are suitable for measurement of single-color LEDs as well as white LEDs.

The conventional sphere-photometers are still acceptable but less preferred due to possible significant spectral mismatch error, especially for measurement of single-color LEDs. Spectral mismatch correction is recommended for measurement of white LEDs and is required for measurement of all single-color LEDs.

Goniophotometers are used when angular intensity distribution is needed. For relative angular intensity distribution measurement, a goniophotometer with a photometer head is sufficient (even for single-color LEDs). Goniophotometers can also be used to measure total luminous flux if they are calibrated for absolute scale and if stray light is carefully controlled; however, the measurement suffers from spectral mismatch errors, and the measurement is time-consuming compared to measurements made with integrating spheres. For measuring both angular intensity distribution and angular color quantity distribution, a goniospectroradiometer or a goniospectroradiometer is needed.

General recommendations for spectroradiometers and spectral measurements are described in ANSI/IES LM-58-20 (see **Section 2.2**). Goniophotometers and goniometric measurements are described in ANSI/IES LM-75-19 (see **Section 2.3**). Integrating spheres and measurement methods are described in ANSI/IES LM-78-20 (see **Section 2.4**).

These instruments shall meet the requirements in the following subsections.

6.1.1 Sphere-Spectroradiometers. A sphere-spectroradiometer shall be equipped with an auxiliary lamp. Self-absorption measurements shall be carried out and a correction made for each product under test, unless the correction factors are found negligible for the type of products to be tested.

The sphere-spectroradiometer should have a proper geometry for the detector port, the DUT, the auxiliary lamp, and the baffle.

The sphere-spectroradiometer shall cover a wavelength range of at least 380 nm to 780 nm for luminous flux and color measurements, and the bandwidth (full-width half-maximum, FWHM) and scanning interval shall be no greater than 5 nm.

The spectroradiometer's input optics at the detector port of the integrating sphere (normally equipped with a diffuser) shall have approximate cosine correction, with an f_2 value of 15% or less.

If the single-pulse mode is used, a sphere-spectroradiometer shall have fast enough response for integration of such short pulses and shall have the capability of making synchronized measurement with the DUT current pulse.

A sphere-spectroradiometer shall be calibrated with a total spectral radiant flux standard traceable to a national metrology institute (NMI). The standards shall be calibrated in appropriate geometries (2π or 4π), depending on the geometry of the sphere system used.

If total spectral radiant flux standard lamps suitable for the small integrating sphere used for the DUT measurement

are not available from the local NMI, a method may be used wherein the relative total spectral radiant flux is calibrated with radiation introduced into the sphere from an external spectral irradiance standard lamp, and the total luminous flux scale can be calibrated against a luminous flux standard source (LED) traceable to an NMI.

6.1.2 Sphere-Photometers. A sphere-photometer should have a total relative spectral responsivity (sphere plus photometer head) that has an f_1' value of 3% or less.

When single-color DUTs are measured, a spectral mismatch correction shall be applied to all measurement results, unless the DUT is compared with the same type of standard LEDs with the same peak wavelength (within ± 5 nm) as that of the DUT.

The photometer head of the sphere-photometer shall have approximate cosine correction, with an f_2 value of 15% or less.

A sphere-photometer shall be equipped with an auxiliary lamp. Self-absorption measurements shall be carried out and corrections made for each product under test unless the correction factors are found negligible for the type of products to be tested.

A sphere-photometer shall be calibrated with total luminous flux standard lamps or LEDs traceable to an NMI. The standards shall be calibrated in appropriate geometries (2π or 4π), depending on the geometry of the sphere system used.

For white LEDs, spectral mismatch correction is not required but is recommended.

6.1.3 Goniophotometers. The photometer head shall have an f_1' value of 3% or less. When single-color DUTs are measured using a goniophotometer with a photometer head, spectral mismatch correction shall be applied to all measurement results, unless the same type of LEDs with the same peak wavelength (within ± 5 nm) as the standard LED traceable to an NMI are measured.

For white LEDs, spectral mismatch correction for a photometer head ($f_1' < 3\%$) is not required but is recommended.

The goniophotometer used for total luminous flux measurement shall be calibrated with a luminous intensity standard or illuminance standard traceable to an NMI, and the measured total luminous flux value (in lumens, lm) shall be verified by measuring the total luminous flux of reference light source or LEDs traceable to an NMI. Alternatively, the goniophotometer may be calibrated against a total luminous flux standard lamp or LEDs traceable to an NMI.

The scanning interval for goniometric measurements depends on the smoothness of the beam profile and beam angle of the DUT. For total flux measurements, the interval required is very dependent on the rate of change in the luminous intensity distribution of the DUT.

For DUTs having a near-Lambertian distribution, the minimum required half-plane interval is 22.5° , and the angle interval within the vertical plane is 5° . DUTs that have rapidly changing luminous intensity distributions may require smaller intervals.

The range of the angular scan of a goniophotometer shall cover the entire solid angle into which the DUT emits light. (Note: Some DUTs have emission over solid angles larger than 2π steradians.)

Care should be taken to minimize stray-light errors. Guidance is available in ANSI/IES LM-75-19 (see **Section 2.3**).

6.1.4 Goniospectroradiometers. Goniospectroradiometers shall be calibrated for spectral irradiance ($\text{W}\cdot\text{sr}^{-1}\cdot\text{m}^{-2}\cdot\text{nm}^{-1}$) or spectral radiant intensity ($\text{W}\cdot\text{sr}^{-1}\cdot\text{nm}^{-1}$) traceable to an NMI.

The spectroradiometer used in the goniospectro-radiometer shall cover the wavelength range of at least 380 nm to 780 nm for photometric quantities and color measurement, and the bandwidth (full-width half-maximum) and scanning interval shall be no greater than 5 nm.

The scanning interval for goniometric measurements depends on the smoothness and beam angle of the DUT. For total flux measurements, the interval required is very dependent on erratic changes in the luminous intensity distribution of the DUT measured.

For DUTs having a near-Lambertian distribution, the minimum required half-plane interval is 22.5°, and the angle interval within the vertical plane is 5°. DUTs that have rapidly changing luminous intensity distributions may require smaller intervals.

Care should be taken to minimize stray light errors. Guidance is available in ANSI/IES LM-75-19 (see **Section 2.3**).

6.2 Optical Measurements

6.2.1 Total Flux. Total luminous flux of the DUT shall be measured using a sphere-spectroradiometer, a sphere-photometer, or a goniophotometer satisfying the requirements described in **Section 6.1**. Total radiant flux of a DUT shall be measured using a sphere-spectroradiometer or a goniospectroradiometer satisfying the requirements described in **Section 6.1**.

6.2.2 Intensity Distribution. Relevant intensity distributions include luminous intensity distribution, radiant intensity distribution, and spectral radiant intensity. The main difference is the instrument required for the goniophotometer. For luminous intensity distribution, a goniophotometer with a photometer head or a goniospectroradiometer is used. Their requirements are described in **Sections 6.1.1** and **6.1.2**. For radiant intensity, a spectroradiometer is needed.

Measurement of LED intensity is described in CIE 225:2017.²

6.2.3 Color Quantities. Color quantities to be measured for a DUT include chromaticity coordinates, (x, y) and/or (u', v') , for all LEDs, and correlated color temperature (CCT), D_{uv} , and general color rendering index (CRI, R_a) for white LEDs. The special CRI (R_i) is optional but is to be reported if required by the customer or regulations. In addition, if requested, color quantities in the latest version of ANSI/IES TM-30—i.e., fidelity index, R_f ; gamut index, R_g ; and other indices—may also be calculated. Color quantities are calculated from the measured relative spectral power distribution (SPD) of the DUT according to the definitions for chromaticity coordinates and CCT, for D_{uv} , and for CRI. Measurement of spectral characteristics for single-color DUTs is described in **Section 6.2.4**.

The color quantities of DUTs shall be measured as values averaged over the light-emitting solid angle. A sphere-spectroradiometer is commonly used for the measurement. The sphere-spectroradiometer shall meet the requirements in **Section 6.1.1**.

The angularly averaged color quantities may also be measured with a goniospectroradiometer (see the requirements in **Section 6.1.4**). For DUTs with a narrow beam distribution, the latitude angle, θ , increments shall be no greater than 1/10 the angle of the beam angle (refer to ANSI/IES LS-1-22; see **Section 2.1**) and no larger than 10°. The color quantities and (relative) luminous intensity at each goniometer angle, which are used for the calculation of color quantities averaged over the light-emitting solid angle, shall be recorded over the angle range where the luminous intensity is more than 10% of the peak intensity.

If necessary, chromaticity uniformity over the emitting angle can be measured using a goniospectroradiometer.

6.2.4 Spectral Characteristics. Spectral characteristics of a single-color DUT include the dominant wavelength, centroid wavelength, and peak wavelength. They are typically calculated from the measured spectral power distribution.

For dominant wavelength, λ_d , the reference achromatic stimulus should be an equi-energy spectrum: a stimulus whose spectral concentration of power as a function of wavelength is constant (sometimes known as Illuminant E) and that has chromaticity coordinates $x_E = 0.3333$, $y_E = 0.3333$.

Peak wavelength, λ_p , is the wavelength at the maximum of the spectral distribution. It shall be measured using a spectroradiometer with bandpass less than 5 nm. The peak wavelength may be determined from the SPD curve interpolated using a function (e.g., cubic spline) and the peak calculated from the function or fine-resolution data.

6.3 Fluorescence Error. Errors can occur when measuring the total luminous flux (lm) of blue LEDs, due to fluorescence from the integrating sphere.

The coating materials themselves generally do not fluoresce, but contamination resulting from, for example, dust particles, can cause fluorescence and is more problematic with older integrating spheres. This is normally not a problem for measuring white light sources; however, the error can be large for blue LEDs because the fluorescence will appear at a longer-wavelength region where the value of $V(\lambda)$ is much higher, and the fluorescence error is magnified. To check such fluorescence of the sphere, the spectrum of a blue LED measured by a sphere-spectroradiometer can be compared with that measured by the directional measurement of the LED spectrum measured outside the sphere (in this case, relative spectral irradiance is measured). It is preferable to use the same spectroradiometer for this comparison (the spectroradiometer is calibrated in each geometry). The difference in the spectral distribution curves at the “wings” of LED spectra in the green region would show the fluorescence, assuming that the blue LED’s spectrum is constant over different angles. Correction can be made by analyzing such data, if necessary.

7.0 Preparation for Measurements

7.1 Seasoning

Seasoning is not required for the methods described in this document. If pre-operation is applied, time and operating conditions before measurement shall be reported.

7.2 Operating Orientation

The light emission process of a DUT is not affected by its orientation. The effect of thermal convection due to different operating orientation of the DUT, if any, is considered negligible with the methods described in this document when the DUT’s T_J is kept constant.

7.3 Thermal Conditions

7.3.1 Junction Temperature, T_J . Optical output and electrical characteristics of a DUT vary with its T_J . Because of this, all DUT electrical and optical measurements made with the methods listed in this

document shall include the T_J value during the optical or electrical measurement.

A T_J of 25 °C is often used as nominal thermal condition; however, any T_J may be used for a measurement to represent the temperature of the DUT operating in an application. For example, a T_J between 50 °C and 100 °C might be used for a DUT that operates at an elevated temperature.

7.3.2 Temperature Control of the DUT. Temperature control of the DUT shall be achieved by using either the ambient air or a TCM:

- **Using the ambient air to control the temperature of the DUT.** For operation in one of the pulse modes, the T_J of a DUT may be set by the ambient air temperature T_A . This method is convenient for the nominal condition (T_J of 25 °C). For elevated ambient temperature conditions, a thermal chamber is required, which could make the total luminous flux measurement difficult.
- **Using a TCM to control the temperature of the DUT.** This method shall be used for a DUT operated under the DC mode, and may be used for a DUT operating in one of the pulse modes. The temperature of the DUT is set by a TCM on which the DUT is mounted. This method should be used for elevated temperatures when one of the pulse modes is used.

7.3.3 Ambient Temperature, T_A . Ambient temperature, T_A , may be used to set the T_J in pulse-mode measurement; conversely, if T_J is controlled by a TCM, the effect of T_A on the DUT’s photometric output is insignificant and T_A may be left uncontrolled, unless the DUT is a remote-phosphor DUT that may be also sensitive to T_A (in addition to its T_J). For measurement of a remote-phosphor DUT, the T_A condition, ± 5 °C from the specified T_J , shall be met.

Generally, the T_A should be measured at a point not more than 0.5 m from the DUT and at the same height as the DUT. The temperature sensor should also be shielded from direct radiation from the DUT.

8.0 Measurement Methods and Procedures

This Approved Method allows DUTs to be operated under a pulsed current or a DC current. The three types of pulse methods, single pulse (SP) methods, flash pulse (FP) methods, and continuous pulse (CP) methods, are all allowed in this document to accommodate various measurement situations. All pulse methods result in a self-heating error (an increase in T_J), which shall be corrected or limited to an acceptable level.

For the SP method, the optical measurement is started after the settling time and finished before the end of the current pulse. The heating of the DUT depends on the settling time before the measurement is made and the total measurement time. The settling time corresponds to the time between switching on the power supply and when the electric current is stabilized. The settling time depends on the current source and DUT. The measurement time is the integration time of the instrument. The minimum measurement time is limited by the minimum integration time of the optical instrument (typically in the range of milliseconds) or the minimum optical signal required. For measurements using the SP method, errors resulting from the heating are significant and therefore shall be corrected. This document provides multiple methods for the correction.

For the FP and CP methods, the DUT is turned on and off within the measurement time window of the optical instrument. Thus, the heating during a pulse can be minimized by reducing the current pulse width, which is limited by the current source. The heating during a pulse is minimal when the pulse duration is a few microseconds.

While short pulses are used in FP and CP methods, the rise and fall errors may become significant due to deviation of the current pulse waveform from an ideal rectangular shape. Therefore, two different FP methods—differential flash pulse (D-FP) method and optical profile flash pulse (P-FP) method—and a differential continuous pulse (DCP) method are described in this document for correcting or minimizing the rise and fall errors. Further,

to eliminate the amplitude errors and timing-related errors (e.g., jitter) in the DCP method, a mean differential continuous pulse (M-DCP) method is described herein.

When the DC method is used, a DUT is operated under a constant DC current, T_J is feedback controlled to the specified temperature, and the entire DUT is in steady state.

Because the various errors in any of the three pulse methods are either corrected or limited to an acceptable level, the measurement results are considered to be equivalent to each other within their measurement uncertainty values. However, the measurement results may not agree between use of a pulse method and use of the DC method for a phosphor-converted DUT (e.g., phosphor-converted white or amber DUTs), because the DUT's phosphor is at a different temperature. Under a pulse operation the phosphor temperature is close to the T_J , while under a DC operation the phosphor temperature is typically tens of degrees higher than the T_J .

For most DUTs, V_F is a suitable parameter for inferring T_J . However, DUTs that exhibit a transient voltage anomaly (thus, not following the Shockley diode equation) shall be excluded, i.e., the near-linear relationship between V_F and T_J is distorted by other factors (e.g., thyristor or snapback effect). For example, a thyristor effect can result in an abnormally high V_F during the first few milliseconds of operation. For these DUTs, relative radiant flux or irradiance should be used to infer T_J . (Refer to **Annex B** for the method for identifying this effect.)

Table 8-1 shows all the methods and their advantages and drawbacks. Any of the measurement methods may be used. When selecting a test method, DUT characteristics, DUT heating, required measurement uncertainty, and allowed measurement time should be considered.

Some characteristics of a DUT can directly rule out some test methods. It may make testing with a method difficult, or it may strongly affect the uncertainty that can be obtained with a method. For example, some laser devices need to be driven in pulsed mode; they

Table 8-1. Comparison of the Measurement Methods

Method / Operation		SP (Single Pulse)			FP (Flash Pulse)		CP (Continuous Pulse)		DC (Direct Current)	
		SP-1	SP-2	SP-3	D-FP	P-FP	DCP	M-DCP		
Source Current to DUT		PCS or P-SMU								
Measure DUT Voltage		DAQ, P-SMU, or DMM								
Sense DUT Current		Not required; relies on the PCS calibration				Current shunt or probe				
Measure Sensed Current		Not required; relies on the PCS calibration				6 ½-digit DMM or better with a multi-slope integrator				
Sense Optical Output for Determining Current Waveform						Not required				
Verify Current Waveform		Required, using an oscilloscope or DAQ								
Control DUT Temperature		Ambient air or TCM							TCM	
Optical Instruments		Photometer, radiometer, or spectroradiometer with hardware triggering capability. The speed of the photometer or radiometer should be at least as fast as the current pulse.			Photometer, radiometer, or spectroradiometer with hardware triggering capability.		Photometer, radiometer, or spectro-radiometer. The integrated dark signal should be small compared to the integrated signal.		Any optical instruments, including goniometric instruments	
Dominant Uncertainty Sources After Correction		5 _{VF} accuracy, and optical-parameter sensitivity variations	Non-linearity of sensitivity factors, especially chromaticity	T _J error due to the error from initial V _F extrapolation	PCS or P-SMU accuracy	Accuracy for optical waveform measurement	PCS or P-SMU accuracy; T _J error due to average temperature rise	T _J error due to average temperature rise	T _J error due to the error from initial VF extrapolation	
Thermal Stabilization Cycle		Initial stabilization is required	Initial and additional stabilizations are required	Total of three stabilizations are required	Initial stabilization is required					Initial stabilization and feedback control are required
Best Use		Rapid measurement of similar DUTs				Rapid measurement of DUTs			Measurement of DUTs under realistic "hot" phosphor condition.	
Usage Not Recommended		DUTs with transient voltage anomaly								DUTs with transient voltage anomaly; DUTs requiring below-dew-point TCM operation

will be destroyed by DC operation. For such DUTs the DC method is not a possibility. Other DUTs can exhibit a voltage anomaly effect that distorts the V_F during the first milliseconds of operation. If the SP method is used to test these DUTs, the temperature corrections that are part of the SP methods shall be made using radiant flux or irradiance (see **Annex B**). For these devices, one of the CP methods is often a better choice, as temperature corrections are not necessary. Conversely, the CP methods may leave a white LED's phosphor at a temperature well below its normal operating point, introducing a measurement error due to increased phosphor conversion efficiency at the lower temperature. Test designers should consider the DUT's characteristics and select a method that best suits the DUT being tested.

DUT heating can create issues during measurements in addition to influencing the optical and electrical parameters being measured. The DC method involves the most heating, as the DUT is powered for long periods. Some DUTs may exceed the capabilities of typical thermoelectric coolers. In addition, some DUTs can heat other items in the test setup, such as spheres and goniometer mounts, to unacceptable temperatures. Even during pulsed operation, heating can become an issue if testing is done at current densities above the DUT's nominal current density. Heat during high-current-density testing can accumulate during an I-V or L-I sweep and can change the resistance of interconnect paths, rendering V_F readings invalid. At very high current levels the DUT can be destroyed. Test designers should assess the average and transient heating and ensure that the DUT, measurement equipment, and uncertainty budget can handle the heating and associated temperature rise.

The DUT's characteristics and heating primarily influence the selection of the test method: SP, FP, CP, or DC. The required measurement uncertainty and allowed measurement time are useful for determining the method variant. For example, SP method variant 1 (SP-1; see **Table 8-1**) is a very quick test, but its temperature correction relies on nominal temperature coefficient of forward voltage SVF and optical-parameter sensitivity factors. Thus, its uncertainty is greater than that of SP-2 or SP-3. However, SP-2 and SP-3 require multiple measurements, and thus more measurement time is

needed. Test designers should assess the measurement uncertainty and the time available and choose a method or method variant that balances the uncertainty budget and the time available.

8.1 The SP Methods

In this method, a single current pulse is applied to the DUT, and an optical measurement is made during the duration of the pulse, after the current has stabilized. SP measurements involve pulses up to tens of milliseconds. For such long pulses, typical current sources can achieve a rectangular shape before the measurement begins and maintain it until the measurement ends. Thus, error sources due to current pulse-profile imperfections do not directly affect the optical measurement. It would seem then, that the rise time of the current pulse is inconsequential, if it completes before the measurements begin. However, the determination of the T_J using graphical projection techniques necessitates a current pulse that rises quickly and settles quickly. Therefore, a current source with a rise time less than 20 μs shall be used. The current pulse width shall also be greater than the integration time (including the interline transfer time of a CCD detector array) of the spectroradiometer used.

The DUT T_J rises continuously during the duration of the current pulse, from its initial nominal temperature to an ending temperature that can be tens of degrees higher. As a result of this rise, the DUT's light output decreases during the pulse. The declining optical output results in an optical measurement error if it is not corrected. The magnitude of the error depends upon the DUT's optical-parameter sensitivity factors. For example, amber AlGaInP LEDs are particularly temperature sensitive. Optical output for these devices can decline more than 10% during a typical single-pulse measurement. Other LED types exhibit much less sensitivity, but output declines can still be in the range of a few percent.

The three SP variants include methods for correcting for the T_J rise during the measurement. The magnitude of the T_J rise is denoted as $\Delta T_{J, \text{rise}}$. These corrections allow the single-pulse measurements to be correlated to the desired reference T_J . **Table 8-2** lists the variants and the method for correction. As discussed above, variants SP-2 and SP-3 rely on inferring the T_J rise using the DUT's

V_F . For most DUTs, V_F is a suitable method for inferring T_J , as the relationship between V_F and T_J is nearly linear over a narrow temperature range. However, some DUTs exhibit non-diode behavior (they do not adhere to the Shockley diode equation). In these DUTs, the near-linear relationship between V_F and T_J is distorted by other factors. For example, in some immature LEDs, a voltage anomaly or snapback effect can produce an abnormally high V_F during the first few milliseconds of operation. For these DUTs, another parameter, such as radiant flux or irradiance, should be used to infer T_J . (Refer to **Annex B** for a method for determining whether a voltage anomaly effect is present.)

For all of the SP methods, electrical measurements shall coincide temporally with optical measurements. In addition, the measurement uncertainty due to triggering variations shall be evaluated and included in the overall measurement uncertainty. (See **Figure 8-1** for a typical setup.)

8.1.1 SP-1 Method. When the SP-1 method is used, a DUT is only measured once. **Figure 8-2** shows timing of a typical single-pulse measurement for total luminous flux. The blue line represents a 30-ms current pulse used to power the device (an AlGaInP LED) during the measurement. The black dashed line illustrates

Table 8-2. The Three SP Methods and Their Correction Approaches

Method	Number of Measurements	Reported T_J	T_J During Measurement	Optical and Electrical Results
SP-1	1	Assign stabilization 1 TCM temperature, T_{TCM} , or ambient temperature, T_A if the ambient air control is used.	Measurement 1 T_{TCM} or T_A + approximate $T_{J,rise}$	Measurement 1 value – (typical temperature coefficient C_T x estimated $\Delta T_{J,rise}$)
SP-2	2	Stabilization 1 T_{TCM}	Measurement 2 T_{TCM} + measured $\Delta T_{J,rise}$	Measurement 2 value – [measured temperature coefficient C_T x (Measure 2 T_{TCM} + measured $\Delta T_{J,rise}$ – Measure 1 T_{TCM}).
SP-3	3	Stabilization 1 T_{TCM}	Measurement 3 T_{TCM} + measured $\Delta T_{J,rise}$	Measurement 3 value

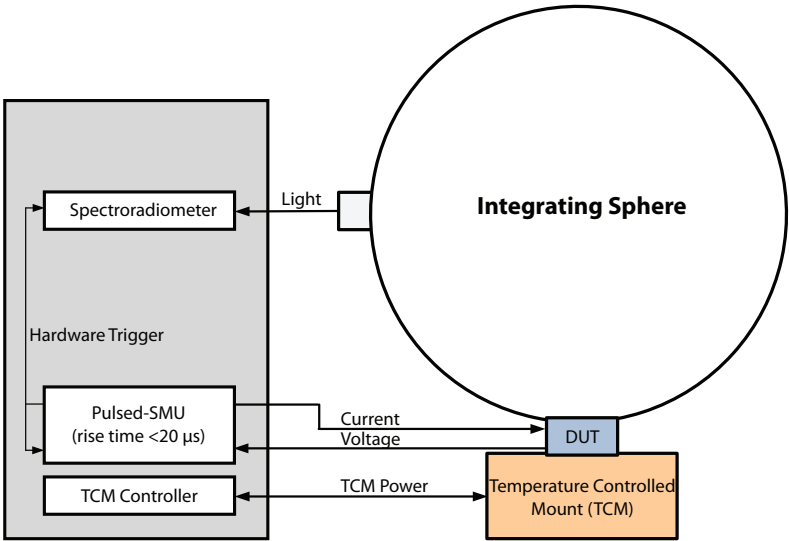


Figure 8-1. Typical SP measurement setup.

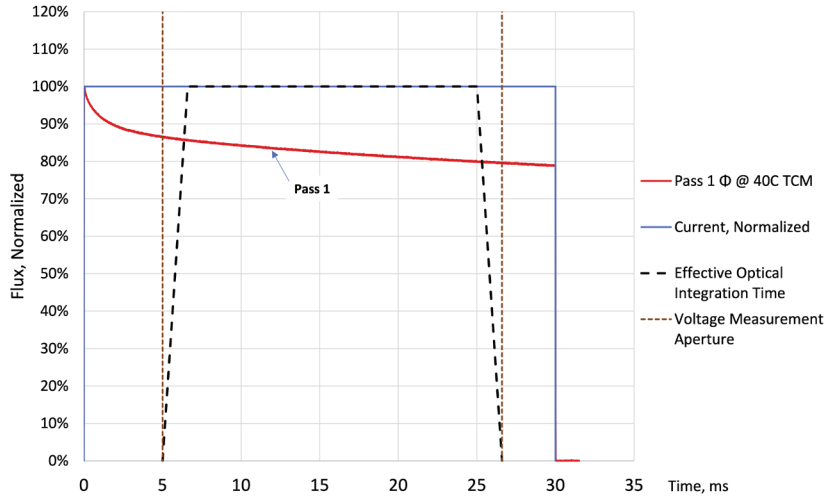


Figure 8-2. SP-1 method's typical measurement timing.

the spectroradiometer integration time, including the rise and fall in sensitivity due to the interline transfer time of its CCD detector. For this nominally 20-ms measurement, the spectrometer begins integrating at 5 ms and ends its integration at 26.5 ms. The LED's flux output is shown by the solid red line. As can be seen from the flux plot, by beginning the measurement at 5 ms, the initial rapid optical decline (from 100% to 85%) is bypassed. The optical measurement is then made from 85% output to roughly 80%. This period of somewhat flatter flux output is the "quasi-stable" period mentioned in the previous version of this standard. Since the spectroradiometer integrates from 5 ms to 26.5 ms, the required voltage measurement aperture is 21.5 ms. Triggers used to coordinate the precision pulsed current source, the spectroradiometer, and the digitizing voltmeter shall be hardware-based to minimize timing variation.

Integrated optical and digitized voltage measurements are collected during the measurement. Measurements are then correlated to the TCM or ambient temperature by estimating $\Delta T_{J,rise}$ during the measurement and applying a correction (based upon the estimated $\Delta T_{J,rise}$) to the resulting optical measurement values. The SP-1 method is ideal for rapid testing of DUTs that have similar, known characteristics.

SP-1 measurements shall be made using the following steps (a flowchart is shown in **Figure 8-3**):

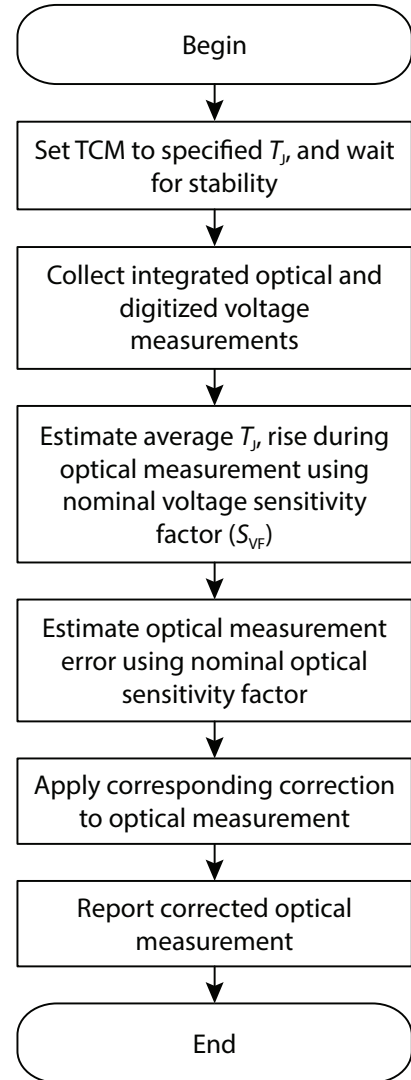


Figure 8-3. SP-1 method flowchart.

1. *Temperature stabilization*: Stabilize the DUT at the specified T_J . (Refer to **Annex A** for a method to determine stability.)
2. *Optical and digitized voltage measurements*: Power the DUT at the test current. Begin optical and electrical measurements after the desired delay, and complete them before the pulse ends. Ensure that electrical and optical measurements coincide temporally.
3. *T_J rise calculation*: Calculate the average V_F during the optical measurement. Also calculate $V_F(0)$, the starting V_F at the moment the DUT was energized. Use extrapolation of V_F samples taken in the first millisecond to bypass any current source electrical

transients. Calculate the average $\Delta T_{J, \text{rise, avg}}$ during measurement as:

$$\Delta T_{J, \text{rise, avg}} = [V_{F, \text{avg}} - V_F(0)] / S_{VF, \text{nominal}} \text{ (for the DUT)} \quad (8-1)$$

The nominal temperature coefficient S_{VF} for the DUT type can be obtained by measuring DUTs of the same type, or it may be obtained from the DUT manufacturer.

4. *Optical-measurement error calculation:* Calculate the estimated optical-measurement error associated with $\Delta T_{J, \text{rise, avg}}$. For an optical parameter, Y , this error is given by:

$$\varepsilon_Y = c_{T_J-Y} \cdot \Delta T_{J, \text{rise, avg}}, \quad (8-2)$$

where the temperature coefficient, c_{T_J-Y} , can be obtained by measuring DUTs of the same type, or it may be obtained from the DUT manufacturer.

5. *Correction of result:* Apply a correction corresponding to the calculated error. Report the corrected optical parameter. Report the TCM temperature, T_{TCM} , or ambient temperature T_A as the reference T_J .

8.1.2 SP-2 Method. In this method, two pulsed measurements are made (see **Figure 8-4**). The two sets of data are then used to calculate the DUT's unique junction temperature coefficient S_{VF} , and the actual $\Delta T_{J, \text{rise, avg}}$, during a measurement. These values are then used to calculate the optical measurement error.

Finally, a corresponding correction is applied to the second optical measurement. The SP-2 method directly measures both $\Delta T_{J, \text{rise, avg}}$ and the optical sensitivity coefficient, c_{T_J-Y} , unique to a particular DUT. The resulting corrections are much more accurate; therefore, the SP-2 method is much more accurate than the SP-1 method.

SP-2 measurements shall be made using the following steps (a flowchart is shown in **Figure 8-5**):

1. *Temperature stabilization:* Set the pass 1 TCM temperature, $T_{TCM}(1)$, to the specified T_J , and stabilize the DUT. (See **Annex A** for a method to determine stability.)
2. *First optical and digitized voltage measurements:* Power the DUT at the test current. Begin optical and electrical measurements after the desired delay, and complete them before the pulse ends. Ensure that electrical and optical measurements coincide temporally (see **Figure 8.4**). Calculate $V_{F, \text{avg}}(1)$, which is the average VF during the first optical measurement.
3. *Temperature adjustment and stabilization:* Reduce the TCM temperature by a fixed amount (e.g., by the estimated $\Delta T_{J, \text{rise, avg}}$ based the DUT type and TCM mounting) to pass 2 TCM temperature $T_{TCM}(2)$. Follow the procedure described in **Annex A** to wait for the thermal equilibrium. Avoid adjusting below the dew point of the ambient air.
4. *Second optical and digitized voltage measurements:* Perform the second measurements. Calculate $V_{F, \text{avg}}(2)$, which is the average V_F during the second

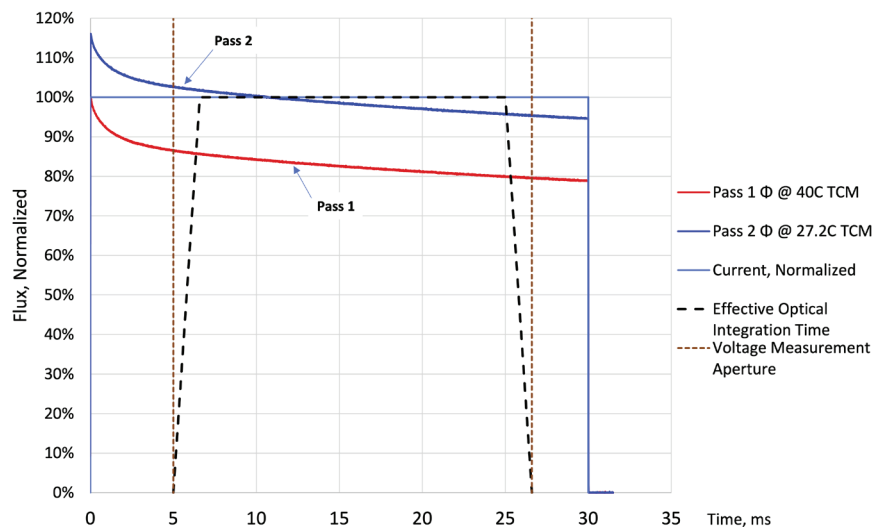


Figure 8-4. SP-2 method's measurement timing.

optical measurement. Also calculate $V_F(0)$, the starting V_F at the moment the DUT was energized. Use extrapolation of V_F samples taken in the first millisecond to bypass any current source electrical transients.

5. *Calculation of temperature coefficient S_{VF}* : The SVF is given by:

$$S_{VF} = \frac{V_{F,avg}(2) - V_{F,avg}(1)}{T_{TCM}(2) - T_{TCM}(1)} \quad (8-3)$$

6. *T_J rise calculation*: Calculate the average T_J rise, $\Delta T_{J,rise,avg}$, during the second measurement as:

$$\Delta T_{J,rise,avg} = \frac{V_{F,avg} - V_F(0)}{S_{VF}(\text{for the DUT})} \quad (8-4)$$

7. *Optical temperature sensitivity and measurement error calculations*: Calculate the temperature sensitivity coefficient for the optical quantity, Y ,

$$C_{TY} = [Y(2) - Y(1)] / [T_{TCM}(2) - T_{TCM}(1)] \quad (8-5)$$

Calculate the estimated measurement error in the second optical measurement as:

$$\epsilon_Y = C_{TY} [T_{TCM}(2) + \Delta T_{J,rise,avg} - T_{TCM}(1)] \quad (8-6)$$

8. *Correction of result*: Apply the correction to the second optical measurement, and report the corrected results. Report the $T_{TCM}(1)$ as the T_J .

8.1.3 SP-3 Method. In this method, three measurement passes are used (see **Figure 8-7**). The first two are used to calculate the DUT's $\Delta T_{J,rise}$ during a measurement. The TCM temperature is then reduced for the third measurement by the amount of the average $\Delta T_{J,rise,avg}$. Thus, the third measurement occurs at the desired T_J , and therefore no measurement corrections are necessary. SP-3 is the most accurate of the single-pulse variants; however, it does require the most time.

SP-3 measurements shall be made using the following steps (a flowchart is shown in **Figure 8-5**):

1. *Temperature stabilization*: Stabilize the DUT at the desired temperature setpoint (refer to **Annex A** for a method to determine stability).
2. *Measurement Pass 1, optical and digitized voltage measurements*: Power the DUT at the test current. Begin optical and electrical measurements after the desired delay, and complete them before the pulse ends. Ensure that electrical and optical measurements coincide temporally (see **Figure 8-6**). Calculate $V_{F,avg}(1)$, the average V_F during the first optical measurement. Also calculate $V_F(0)$, the starting V_F at the moment the DUT was energized. Use extrapolation of V_F samples taken in the first millisecond to bypass any current source electrical transients.

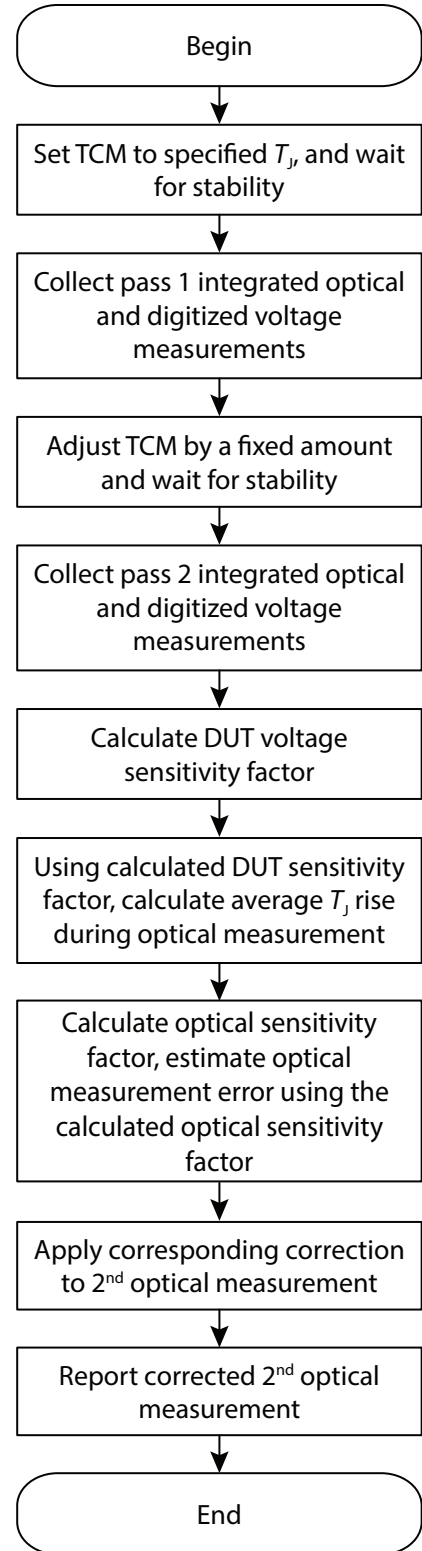


Figure 8-5. SP-2 method flowchart.

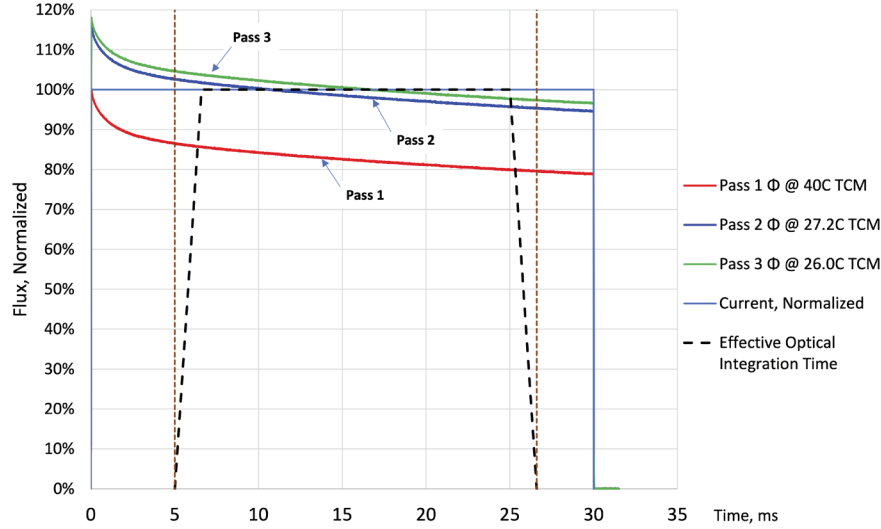


Figure 8-6. SP-3 method's measurement timing.

3. *Temperature adjustment and stabilization:* Adjust the temperature of the TCM, T_{TCM} , by a fixed amount (e.g., by 1 °C or 10 °C); avoid adjusting below the dew point of the ambient air. Follow the procedure described in **Annex A** to wait for the thermal equilibrium.
4. *Measurement Pass 2, optical and digitized voltage measurements:* Repeat the Pass 1 measurements. Calculate $V_{F,avg}(2)$, the average V_F during the second optical measurement. Also calculate $V_F(0)$, the starting V_F at the moment the DUT was energized. Use extrapolation of V_F samples taken in the first millisecond to bypass any current source electrical transients.
5. *Calculation of the temperature coefficient, S_{VF} :* The S_{VF} is given by:

$$S_{VF} = \frac{V_{F,avg}(2) - V_{F,avg}(1)}{T_{TCM}(2) - T_{TCM}(1)} \quad (8-7)$$

6. *T_J rise calculation:* Calculate the average $\Delta T_{J,rise,avg}$ during the first or second measurement as:

$$\Delta T_{J,rise,avg} = \frac{V_{F,avg} - V_F(0)}{S_{VF}(\text{calculated for the DUT})} \quad (8-8)$$

7. *Temperature adjustment:* Adjust T_{TCM} below the nominal DUT measurement temperature by an amount equal to the calculated $\Delta T_{J,rise,avg}$ during the optical measurement.
8. *Measurement Pass 3, optical and electrical measure-*

ments: Make the third-pass optical measurement at the reduced TCM temperature. The average T_J during the measurement now equals the nominal measurement temperature.

9. *Reporting:* Report the third optical measurement, the Pass-1 TCM temperature as the reported T_J , and the TCM temperature reduction that was made for the Pass-3 measurements.

8.2 Flash Pulse (FP) Method

The FP measurement methods also use a single-pulse operation. However, the pulse falls into the optical measurement window. Compared to the SP methods, FP methods use a much shorter current pulse to minimize $\Delta T_{J,rise}$. The maximum pulse width shall not be greater than 50 μ s. When a pulse width greater than 50 μ s is used, the measurement result should be corrected for the $\Delta T_{J,rise}$. This short pulse may lead to a non-rectangular pulse waveform and therefore results in rise and fall measurement errors. To reduce or eliminate the timing error of the pulse and the rise and fall errors, one of two FP approaches, the differential FP method (D-FP method) or the optical profile FP method (P-FP method), shall be applied.

8.2.1 D-FP Method. When a D-FP method is used, the DUT is operated, in turn, using two narrow (<50 μ s) current pulses, short single pulse (P_S) and long single pulse (P_L), and is measured for optical and electrical values. The two measured optical results are subtracted to obtain a result equivalent to that when the DUT is

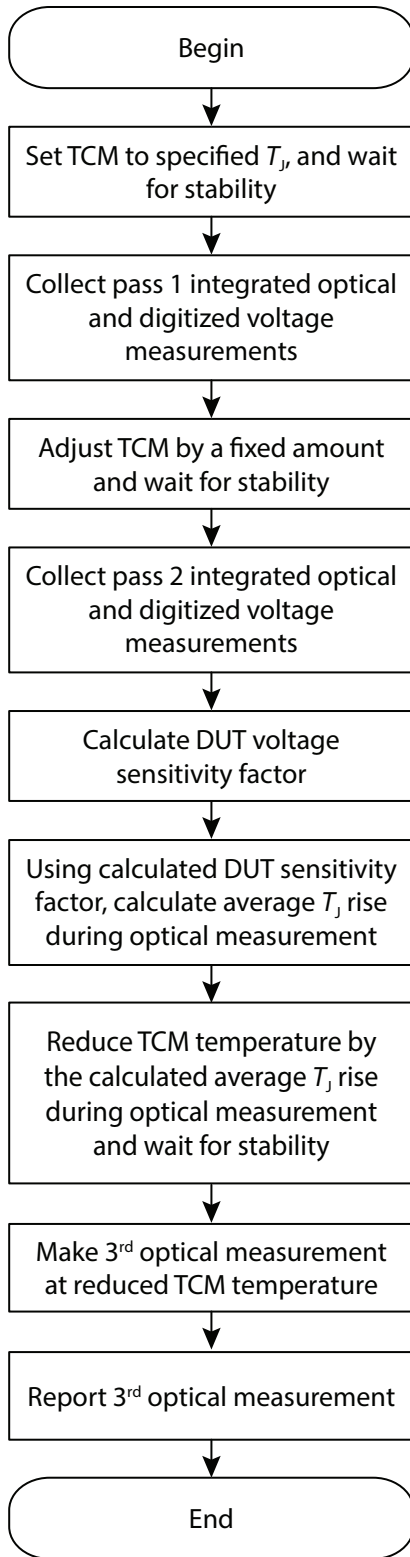


Figure 8-7. Variant SP-3 flowchart.

operated with a square current pulse. The principle of this method is illustrated in **Figure 8-8**.

Care should be taken that the optical and electrical sensors and instruments are not saturated at the peak value.

8.2.1.1 Ambient Temperature Condition. The requirements in **Section 4** also apply for the D-FP method.

8.2.1.2 Optical and Electrical Measurement Procedures. The measurement shall follow the procedures described in the steps below.

1. *Temperature stabilization.* Stabilize the temperature of the DUT to the desired T_j via one of these methods:
 - *When using the ambient temperature control:* For the room temperature condition or elevated temperature condition using a thermal chamber, the ambient temperature or thermal chamber air temperature shall be set to the desired T_j .
 - *When using a TCM:* The DUT is mounted on the TCM, the TCM temperature, T_{TCM} , shall be set to the specified T_j and the DUT T_j allowed to stabilize. (Refer to the methods in **Annex A** to determine whether the DUT is stable.)
2. *Starting integration of optical instrument.* Wait for the instrument's responsivity to reach the maximum value.
3. *Applying the long current pulse P_L and measuring the voltage.* Hardware-trigger the current source to apply the long current pulse, P_L , to the DUT. Measure the V_f at the plateau region of P_L .

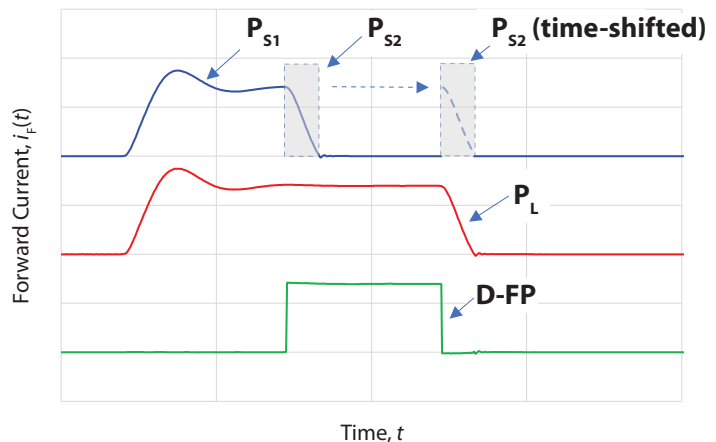


Figure 8-8. Principle of the D-FP method.

4. *End of integration and electrical measurement.* Calculate the measured optical signal, Y_{P_L} , for the long pulse, P_L .

5. *Repeating Step 2 to Step 4 for the short current pulse, P_S .* Obtain the optical signal, Y_{P_S} , under the P_S condition. No need to measure V_F under P_S .

6. *Calculating the differential optical signal, Y_{D-FP} .*

$$Y_{D-FP} = Y_{P_L} - Y_{P_S}$$

7. *Converting to DC-operation values, Y_{DC} .* Convert the measured optical value, Y_{D-FP} , to the value when the DUT is operated under DC, Y_{DC} , using:

$$Y_{DC} = Y_{D-FP} \times \frac{t_{\text{Integration time}}}{t_{D-FP \text{ pulse width}}} \quad (8-9)$$

8.2.2 P-FP Method. The electrical pulse shape is not perfectly rectangular, even if high-end instruments are used; consequently, the optical pulse shape is not perfectly rectangular. If the optical measurement is taken using the FP (or CP) mode with the optical measurement time greater than the optical pulse width (e.g., a spectroradiometer measurement), the result is influenced by these deviations from the ideal rectangular shape and a correction is needed. **Figure 8-9** illustrates the principle of the P-FP method that corrects the measurement error from an imperfect rectangular waveform. The P-FP method enables the determination of the correction of the optical signal to the nominal pulse width of a rectangular pulse of the same width using a single measurement.

The correction method is also applicable to multi-pulse methods if the shape of the pulses is constant.

This correction does not include a correction for DUT self-heating effects. Self-heating effects depend on the width of the electrical pulse and can be reduced by shortening the electrical pulse width.

If the voltage and optical measurements cannot be taken from the same electrical pulse, the voltage measurement shall be taken before the optical measurement.

Since this approach uses an optical measurement rather than a current measurement, it implicitly includes the correction of optical errors (the rise and fall errors) as well as amplitude and timing errors. This approach requires a time-sampled signal and a fast-responding photodiode, which is approximately spectrally flat in responsivity. The following steps shall be used:

1. Measure the time-resolved signal of the optical quantity with a nominal pulse width, t_p .
2. Determine the area corresponding to the complete measured pulse area of the single pulse, A_{p-m} .
3. Determine the area, A_{StS} , for which the signal sequence is linear and its value or its slope is constant during a time t_{StS} .
4. Calculate the pulse-profile correction factor, F_{pp} :

$$F_{pp} = \frac{A_{StS}}{A_{p-m}} \cdot \frac{t_p}{t_{StS}} \quad (8-10)$$

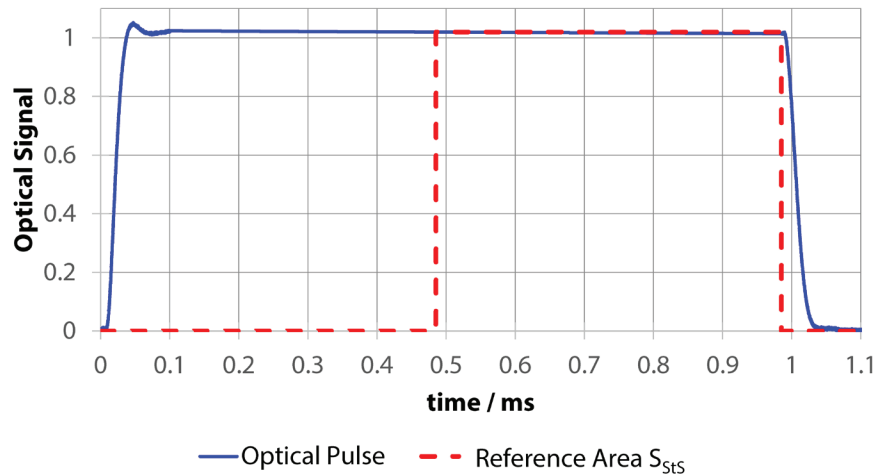


Figure 8-9. Optical pulse (amber) and reference area (blue, A_{StS}) of width t_{StS} used for the correction.

5. Determine the factor correcting the difference of the pulse width and the integration time, F_{int} :

$$F_{\text{int}} = t_{\text{int}}/t_p, \quad (8-11)$$

where:

t_{int} = integration time of the optical detector

6. The optical pulse is captured using a fast photodiode, and the measured signal is then digitized using an oscilloscope or a fast digitizer. The resolution of the analog-to-digital conversion shall be greater than 10 bits.

7. The corrected optical quantity, y , then is calculated from:

$$y = F_{\text{int}} \cdot F_{\text{pp}} \cdot y_{\text{meas}}, \quad (8-12)$$

where:

y = quantity, corrected for pulse profile and

$$t_{\text{int}} > t_p$$

y_{meas} = measured quantity without correction (measurement data)

The F_{pp} in Step 4 is calculated from the quotient of two integral values. Each of the values is retrieved by temporal integration of a photodiode signal proportional to the pulsed radiant emission of a light source. The uncertainty of the F_{pp} is influenced by the measurement system (photodiode, amplifier, DAQ). The following equipment properties do affect the photodiode signal quality:

- *Photodiode rise time.* The photodiode rise time shall be significantly shorter (by a factor of 10) than the rise time of the measured signal.
- *Photodiode responsivity.* The photodiode should have a linear responsivity within the measured signal.
- *Signal amplifier gain.* The signal amplifier gain should be adequate for the measured photodiode current and the data acquisition unit. It is recommended that a signal amplifier be used that provides the functionality of setting a variable gain, in order to have the capability of measuring light sources with different levels of optical power.
- *Photodiode wavelength range.* The photodiode wavelength range should be selected based on the spectral distribution of the light source.

- *Signal amplifier rise/fall time.* The signal amplifier rise/fall time shall be sufficiently short (<100 ns) to display the rising and falling edge of the photodiode signal.
- *Data acquisition bandwidth.* The bandwidth of the DAQ should be sufficiently large for resolving the measured signal with the required accuracy.

8.3 Continuous-Pulse (CP) Methods

In the continuous-pulse method, narrow (microseconds in length) continuous current pulses with a small duty cycle, (e.g., 1%) at a specific frequency are applied to the DUT, and an optical measurement is made for the time-averaged signal, similar to a steady DC measurement.

Figure 8-10 shows an example of current pulses applied and the sequence of measurement in this method. The duty cycle, D , is then given by:

$$D = t_{\text{on}} / (t_{\text{on}} + t_{\text{off}}), \quad (8-13)$$

where:

t_{on} = the duration of each current pulse

t_{off} = the time between pulses

With this method, the pulse width and duty cycle should be small enough that optical measurement errors due to the rise in T_j are within acceptable measurement uncertainty. Short pulses ensure that the transient heating is minimized, and average heating is reduced by the reciprocal of the duty cycle. For most DUTs, the combined T_j rise is then small enough to be considered insignificant, since a mechanism for preventing slow heat buildup, such as a TCM or active airflow, is present. This is the primary advantage of this method.

While the CP method can eliminate heating errors, it introduces some key requirements on the test instruments.

Due to the low duty cycle of the pulse (e.g., 1%), the instrument's detector receives high peak signals compared to the measured integrated signal. Special care should be taken that the instruments are not saturated with the high peak emission of the pulsed DUT. In addition, since the instrument is integrating through both the t_{on} and t_{off} times, it is critical that instrument have a low dark signal. This is especially true for measurements that involve small signal levels—for example, low-current measurements within an L - I sweep.

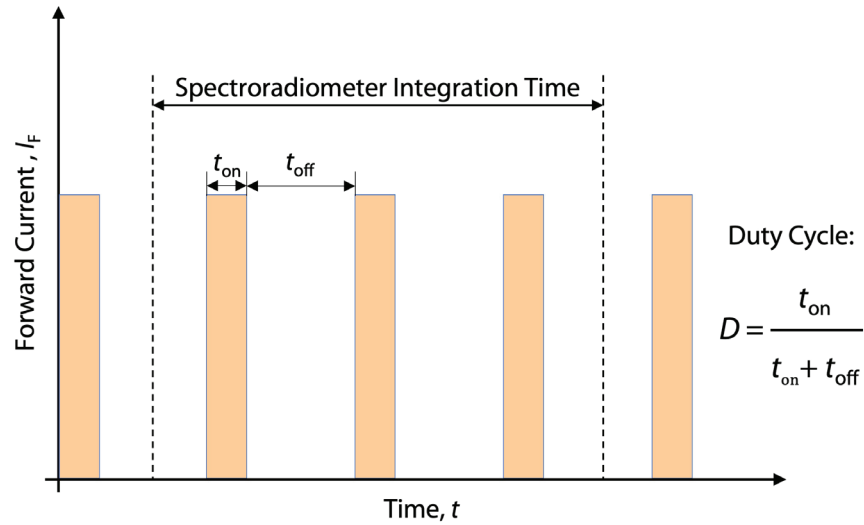


Figure 8-10. An example of measurement sequence in continuous-pulse (CP) mode.

In principle, for spectroradiometers calibrated with steady DC light, the measured photometric output of the DUT is divided by the duty cycle of the current waveform to obtain the final value; thus, the accuracy of the pulsed-current source duty cycle is critical to reduce duty cycle errors, especially when the current pulse waveform is very short and not perfectly rectangular.

The primary continuous-pulse error sources are:

- *Pulse width and period imperfections.* These cause duty cycle factor to be incorrect.
- *Pulse amplitude distortions.* Tilt, sag, overshoot, and settling time cause assumed average current amplitude to be incorrect.
- *Pulse rise and fall inclusion.* Proportionally more light is produced when the DUT is driven at less than the nominal current level, due to increased quantum efficiency. This proportional error is generally positive.

To eliminate or minimize these errors, one of the two approaches shall be used to make optical and electrical measurements: 1) differential continuous pulse (DCP); or 2) mean differential continuous pulse (M-DCP).

8.3.1 Differential Continuous Pulse (DCP) Method.

In this approach, two separate continuous-pulse train measurements are made using two different pulse widths and duty cycles. The first pulse, P_S , is a shorter pulse that includes mainly rise and fall sections. The second pulse, P_L , is a longer pulse that has a nearly flat section in addition to the rise time and sections. The measured signal from the second measurement is then subtracted from that of the first. The resulting differential signal represents the light output corresponding to the flat portion of the longer pulse, P_L , excluding the DUT's current rise time and fall time. The method is shown graphically in **Figure 8-11**.

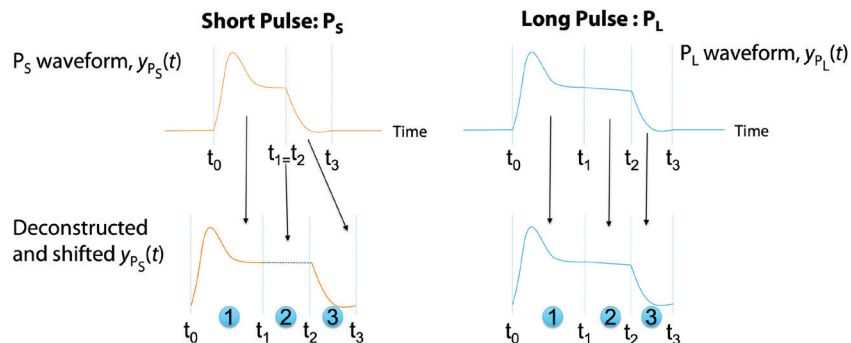


Figure 8-11. Principle of the DCP method.

Two continuous-pulse measurements are used in this method. The first measurement utilizes a shorter current pulse, P_S , (e.g., 10 μs , or 20 μs) than the second measurement, P_L (e.g., 20 μs , or 50 μs). Both measurements use the same pulse period.

A current pulse is composed of three regions:

1. Rising and settling
2. Stable operation
3. Falling

For most accurate pulsed-current sources, the rising and settling characteristic is the same regardless of pulse width. The same is true for the falling characteristic. Therefore, when the optical signal of the second CP measurement is subtracted from the optical signal of the first, the contribution to the optical signal from the rising and settling region and that from the falling region cancel out. The remaining optical signal is from the flat region of the second current pulse: stable operation. In this way, the DCP method removes pulse amplitude distortions.

Current pulse-width imperfections are sometimes similar for two pulses generated by the same PCS. For example, if the source has an error of +1 μs at 20 μs , it may exhibit the same error at 10 μs . If this is the case, then pulse-width errors cancel out.

Pulse-period imperfections may or may not be similar for the first and second measurements. Even if they are similar, the error is not removed by the subtraction.

Most of the error due to the dark signal (the portion corresponding to the off time of the first measurement) is removed by the subtraction.

Thus, the DCP method removes many of the errors associated with the traditional CP technique.

For integrated optical measurements, such as radiant-flux measurements, acceptable results can be obtained by subtracting the measured flux values. However, for non-integrated measurements, such as peak wavelength, direct subtraction is not valid. For non-integrated measurements, the respective P_S and P_L

spectral power distributions (SPDs) shall be subtracted first, and then the parameter of interest shall be calculated from the resulting difference SPD.

Using the DCP method, the measurement equations for the optical signal are as follows:

As shown in **Figure 8-11**, for an individual pulse, the time-integrated total optical signal, Y , generated by P_S is:

$$Y_{P_S} = \int_{t_0}^{t_3} y_{P_S}(t)dt = Y_{P_S, t_0 \rightarrow t_1} + Y_{P_S, t_1 \rightarrow t_2} + Y_{P_S, t_2 \rightarrow t_3} \quad (8-14)$$

For the short pulse, $t_1 = t_2$. Thus,

$$Y_{P_S, t_1 \rightarrow t_2} \equiv 0 \quad (8-15)$$

The time-integrated total optical signal, Y , generated by P_L is

$$Y_{P_L} = \int_{t_0}^{t_3} y_{P_L}(t)dt = Y_{P_L, t_0 \rightarrow t_1} + Y_{P_L, t_1 \rightarrow t_2} + Y_{P_L, t_2 \rightarrow t_3} \quad (8-16)$$

Assuming that:

$$Y_{P_L, t_0 \rightarrow t_1} + Y_{P_S, t_0 \rightarrow t_1} \quad (8-17)$$

the measured DCP optical signal, Y_{DCP} , is:

$$Y_{DCP} = Y_{P_L} - Y_{P_S} = Y_{P_L, t_1 \rightarrow t_2} + [Y_{P_L, t_2 \rightarrow t_3} - Y_{P_S, t_2 \rightarrow t_3}] \quad (8-18)$$

The first term resulting from the subtraction, $Y_{P_L, t_1 \rightarrow t_2}$, represents a trapezoidal area with vertical sides and anear flat top.

The second term, $[Y_{P_L, t_2 \rightarrow t_3} - Y_{P_S, t_2 \rightarrow t_3}]$, is a negligible value that is actually negative because the P_L optical signal at t_2 is less than the P_S signal at t_2 .

Thus, the differential optical measurement equation is:

$$Y_{DCP} = Y_{P_L} - Y_{P_S} \approx Y_{P_L, t_1 \rightarrow t_2} \quad (8-19)$$

While the differential correction method requires two separate CP measurements, P_S and P_L , it provides correction of many of the primary CP error sources:

- *Pulse-width and period imperfections.* Pulse-width errors are generally consistent for both pulse widths, and so they cancel out. For typical current

sources with hardware-based timing, pulse-width error is reduced by 2 to 3 orders of magnitude.

- *Pulse-amplitude distortions.* Tilt, sag, overshoot, and settling-time distortions are generally confined to the beginning of the pulse. Through appropriate choice of pulse width, they will also be eliminated.
- *Rise and fall inclusion.* The rising and falling edges for both current pulse widths are usually identical; therefore, the optical measurement errors due to the edges should cancel out. In reality, the falling edge of the longer pulse produces slightly less light due to higher T_J , but with pulses under 50 μs , the error is generally negligible.
- *Dark signal.* Most of the error due to the dark signal (the portion corresponding to the off time of the second measurement) is also removed by the subtraction.

The duty cycle shall not exceed 2% in order to limit the DUT heating. The width of PL should be as short as possible to minimize DUT heating. DUT heating is generally insignificant when $P_S = 10 \mu\text{s}$, $P_L = 20 \mu\text{s}$, and a duty cycle of 1% are used.

Important: A P_L width greater than 50 μs shall not be used unless the DUT heating is negligible, such as during operation under low current.

The width of P_S used for this method should be chosen to achieve the smallest measurement uncertainty by maximizing the dynamic range. Therefore, the shortest possible P_S width is desired. However, the P_S cannot be shorter than the sum of the current source timing and amplitude errors. That is, P_S width shall be larger than the total of: rise time + settling time + pulse-width error + pulse-width jitter + fall time.

The following steps shall be followed to make DCP measurements:

1. *Temperature stabilization.* One of the following methods shall be used:
 - *When using the ambient temperature control:* For the room temperature condition or an elevated temperature condition using a thermal chamber, the ambient temperature or thermal chamber air temperature shall be set

to the desired T_J . If the DUT exhibits a stable V_F characteristic, stabilization of the DUT's T_J may be monitored by applying a small monitoring DC current, I_M , (e.g., 1 mA) through the DUT while monitoring the DC or pulsed voltage, V_M . When the measured V_M is stabilized within the range determined by the DUT's S_{VF} and the measured parameter's temperature sensitivity factor, the DUT is considered stabilized. If the temperature of the DUT at its thermal contact point (pin, case, or board) is monitored, it is considered stabilized if the monitored temperature remains within $\pm 0.5^\circ\text{C}$ of the specified T_J for at least 5 minutes.

- *When using a TCM:* The DUT is mounted on the TCM, the TCM temperature is set to the specified T_J , and the TCM temperature, T_{TCM} , is allowed to reach and stabilizes to the set temperature. The T_{TCM} is considered stabilized if it remains within $\pm 0.2^\circ\text{C}$ of the specified T_J for at least 1 minute.

2. *Applying the continuous-pulse currents (P_S and P_L).* After programming the PCS to generate the required P_S or P_L waveform, wait for the time required for the PCS current output to stabilize.

3. *Optical and electrical measurements.* Perform optical measurements for a specified duration of integration time for a time-averaged signal. If the measurement is not synchronized with the pulse period, the integration time shall be exactly an integer multiple of the period of the pulses, to avoid errors due to the timing of the integration. The measured signal should be more than 10% of the instrument's full range.

4. *Performing the subtraction.* Subtract the P_S optical signal from the P_L optical signal to obtain the DCP optical signals. Scale the resulting DCP optical signal, Y_{DCP} , to the DC signal, Y_{DC} , corresponding to that under DC operation, using the formula:

$$Y_{DC} = \frac{Y_{DCP}}{D}, \quad (8-20)$$

where D is the duty cycle of the DCP.

5. *Measure the DUT V_F .* The DUT V_F is an important parameter that should be measured whenever an optical measurement is made. For DUTs with no transient-voltage anomaly effect, the average

V_F should be measured during the second half of the DCP long pulse. This shall be accomplished using a DMM (or a digitizer) that is triggered with a hardware trigger signal from the PCS. This hardware triggering ensures that the average measurement corresponds to the optical measurement time (t_1 to t_2 of pulse P_L). The DMM shall be configured to have a trigger delay time (t_0 to t_1 of pulse P_L) and a measurement aperture time that matches the optical measurement time (t_1 to t_2 of pulse P_L). For DUTs with a transient-voltage anomaly effect, V_F of the DUT shall be measured under DC operation with a TCM temperature so that the measured peak wavelength, radiant flux, or irradiance is equal to that measured under the DCP operation.

8.3.2 Mean Differential Continuous Pulse (M-DCP)

Method. The M-DCP method improves on the DCP method by eliminating the following errors due to an imperfect current source and circuitry³:

- Current pulse-width errors
- Current pulse-period errors
- Current pulse-amplitude distortions during the flat part of the pulse

These errors are eliminated by measuring the mean current and/or signal amplitude for both the P_S and the P_L waveform, using a DMM with a multi-slope integrator or a high accuracy digitizer. The mean current is then used to calculate the exact duty cycle of the current pulse train. The measured duty cycle is used

to convert all the measured optical mean values to the corresponding values under DC operation with a specified operating current. The M-DCP method should be selected whenever the measurement uncertainty due to P-SMU timing or amplitude errors exceeds the uncertainty budget. For example, **Table 8-3** illustrates a typical uncertainty calculation for a 10- μ s/20- μ s, 1%/2% DCP measurement.

For the P-SMU in this example, the optical measurement uncertainty associated with the P-SMU timing errors is about 3.4%. Much of this uncertainty could be removed by using the M-DCP method. The principle of this method is illustrated in **Figure 8-12**.

An instrument setup for the M-DCP method is shown in **Figure 8-13**. It is important to note that DUT current is sensed with a current probe or with a shunt. The signal from this sensor is then measured using a DMM with a multi-slope integrator.

Other than the current measurement, the M-DCP method is like the DCP method. The selection of P_S and P_L shall be made as in the DCP method.

The following procedures shall be followed to make M-DCP measurements:

1. *Temperature stabilization.* Perform temperature stabilization as in the DCP method.
2. *Applying the continuous current pulse, P_S .* Apply the short pulsed-current waveform as described in the DCP method.

Table 8-3. Example Calculation of PCS Errors

PCS Characteristic	Specified Accuracy	Contribution to Relative Standard Uncertainty
Pulse Width Accuracy, P_S	0.2 μ s	1%*
Pulse Width Accuracy, P_L	0.2 μ s	0.5%*
Pulse Width Jitter, P_S	0.11 μ s	1.1%
Pulse Width Jitter, P_L	0.11 μ s	0.55%
Pulse Period Jitter, P_S	2 μ s	0.2%
Pulse Period Jitter, P_L	2 μ s	0.2%
Relative Combined Uncertainty		1.7 %
Relative expanded Uncertainty ($K = 2$)		3.4 %

* Table note: Uncertainty calculation assumes one-half of SMU pulse-width error is common to both P_S and P_L .

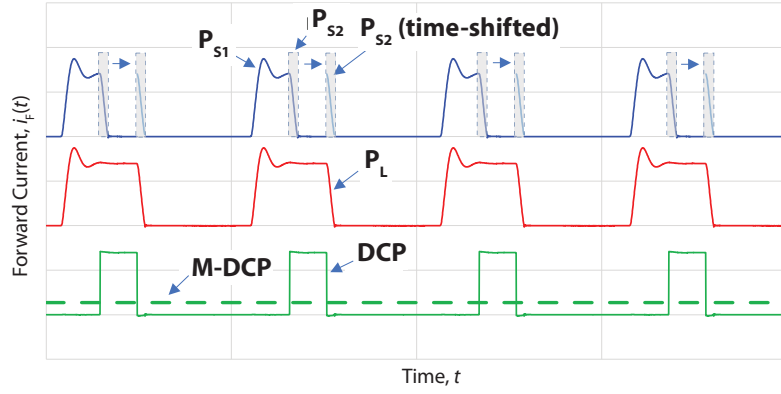


Figure 8-12. Principle of the M-DCP method.

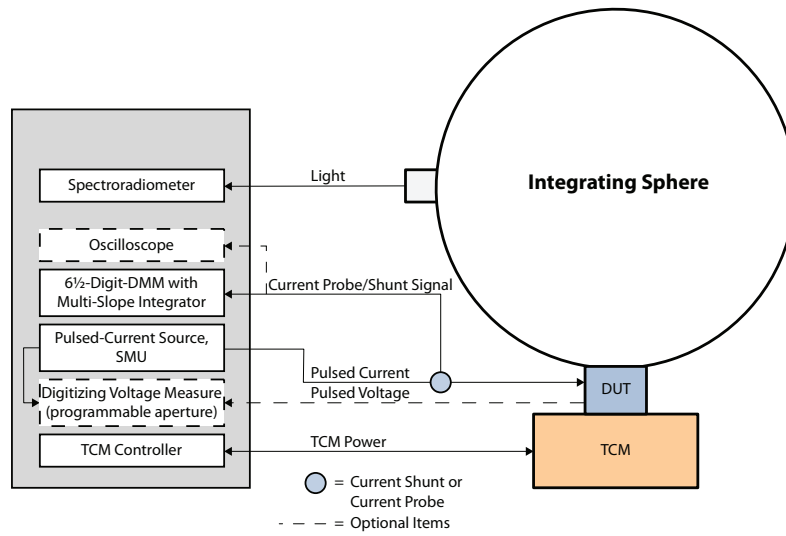


Figure 8-13. Measurement setup for M-DCP method.

3. *Optical and electrical measurement.* Measure the mean optical signal, $y_{P_S, \text{mean}}$, under the P_S condition. Also, measure the mean current, $I_{F, P_S, \text{mean}}$, of the P_S waveform using a DMM with multi-slope integrator or high-speed DAQ.

4. Repeat Step 2 and Step 3 under the P_L condition. Obtain the $y_{P_L, \text{mean}}$ and $I_{F, P_L, \text{mean}}$ under the P_L condition.

5. *Calculation of the mean values.* Calculate the differential mean values using:

$$y_{\text{DCP, mean}} = y_{P_L, \text{mean}} - y_{P_S, \text{mean}} \quad (8-21)$$

$$I_{F, \text{DCP, mean}} = I_{F, P_L, \text{mean}} - I_{F, P_S, \text{mean}} \quad (8-22)$$

6. *Conversion to DC operation values.* Convert the differential mean value, $y_{\text{DCP, mean}}$, to the value y_{DC} when the DUT is operated under the specified DC current, I_F , using:

$$y_{\text{DC}} = y_{\text{DCP, mean}} \times \frac{I_F}{I_{F, \text{DCP, mean}}} \quad (8-23)$$

7. *Measurement of the DUT V_F .* Measure the DUT V_F as in the DCP method described in Step 5 of **Section 8.3.1**.

8.4 DC Method

8.4.1 General. The DC method controls the T_J of a DUT to a specified constant temperature for optical and electrical measurements under steady DC operation.⁴ Pulse operation is required, but it is only used for digitizing the V_F curve due to T_J rise, to set the DC operating condition of the DUT. This method requires a TCM, a fast-current source, and a fast DMM (or digitizer).

For most DUTs, it is suitable to infer T_J from V_F . However, DUTs that exhibit a transient voltage anomaly (the Shockley diode equation is not followed) shall be excluded. The near-linear relationship between V_F and T_J may be distorted by other factors. For example, the thyristor effect can result in an abnormally high V_F during the first few milliseconds of operation. For these DUTs, peak wavelength, radiant flux, irradiance, or another optical parameter that is sensitive to T_J shall be used to infer T_J . (Refer to **Annex B** for identifying the transient voltage anomaly effect.)

8.4.2 Ambient Temperature. Ambient temperature may affect measurement results (especially when measuring remote-phosphor LED sources), depending on the DUT. This is mainly due to the influence on the temperatures of the phosphor and other optical materials of the DUT. The room temperature is typically used as the ambient temperature, T_A , and shall be measured and reported.

8.4.3 Measurement Procedure. The following procedure shall be followed to set the T_J of the DUT.

1. *Mounting the DUT.* Mount a DUT on a TCM by using a metal-core printed circuit board (MCPCB) or any similar mounting technique.
2. *Setting the TCM temperature.* Set the temperature of the TCM to the desired T_J .

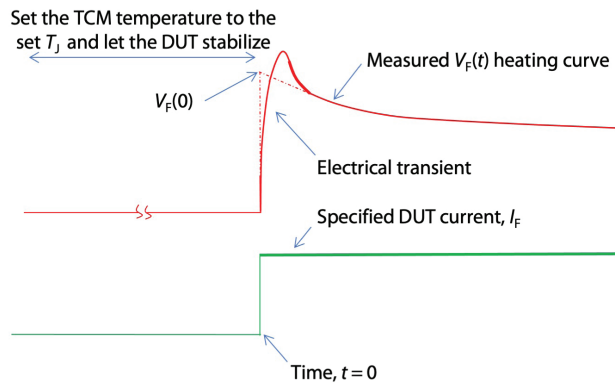


Figure 8-14 (a). Illustration of the measurement procedure: extrapolating $V_F(t)$ for obtaining $V_F(0)$.

3. *Stabilizing the DUT.* Wait until the DUT T_J is stable. (See the methods in **Annex A** to determine whether the DUT is stable.)

4. *Determining $V_F(0)$.* Apply a short current pulse (e.g., 10 ms) with the rated current, and immediately measure the instantaneous forward voltage curve, $V_F(t)$. Determine $V_F(t)$ at $t = 0$, $V_F(0)$, by extrapolating the V_F samples taken in the first millisecond to bypass any electrical transients [see the illustration in **Figure 8-14 (a)**]. If necessary, repeat the measurement to obtain an averaged $V_F(0)$ for higher accuracy.

If the V_F is already obtained using a DCP or M-DCP method for the desired T_J (see Step 5 in **Section 8.3.1**), then steps 2 to 4 are not needed.

5. *Turning on the DUT at DC mode.* Lower the TCM to the estimated required temperature, and then turn on the DUT at the specified DC current. Adjust the TCM temperature so that the measured V_F is equal to the $V_F(0)$ obtained in Step 4, indicating that T_J is the same as when it was turned on at $t = 0$. (Note: This TCM temperature adjustment can be automated.) [See the illustration in **Figure 8-14 (b)**.]

6. *Optical and electrical measurements.* Measure any optical (radiometric, photometric, and/or colorimetric) quantities of the DUT under the DC condition. The results obtained are the values at the specified T_J .

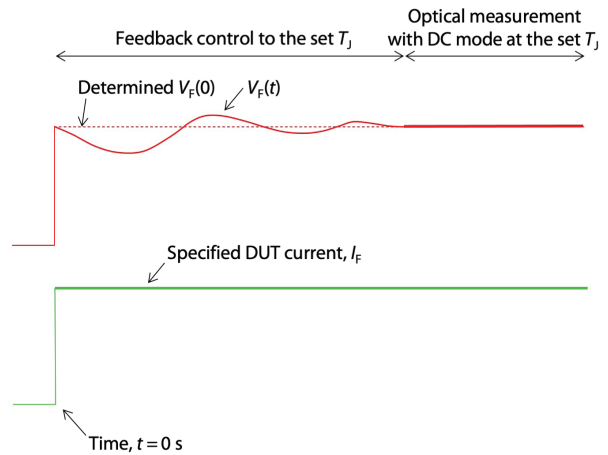


Figure 8-14 (b). Illustration of the measurement procedure: turning on the DUT and feedback-control TCM temperature for $V_F(t) = V_F(0)$.

9.0 Measurement Uncertainty

If a statement of uncertainty is required, the recommendations given in ISO/IEC Guide 98-3:2008 (see **Section 2.5**) and JCGM 104:2009 (see **Section 2.6**) shall be followed. For all photometric and electrical measurements, expanded uncertainty with a confidence interval of 95% shall be used, thus in most cases using the coverage factor $k = 2$.

To verify the measurement uncertainty, the measurement results of some DUTs can be compared with measurements traceable to an NMI.

10.0 Test Report

The report shall list all pertinent data concerning the conditions of testing. The report shall also list the type of equipment, DUT, and reference standards used. Typical items reported are:

- Date of measurement and testing agency
- Manufacturer's name and the designation of the DUT
- Measured quantities; for example, total luminous flux, luminous efficacy, total radiant flux, radiant efficiency, and chromaticity

- Specified values: electrical input power and I_F ; nominal CCT (for white LEDs)
- Method of measurement: single pulse, continuous pulse, DC mode, pulse width (for pulse modes), and duty cycle (for continuous-pulse mode)
- T_J of the DUT, and ambient temperature, T_A
- Instrument and geometry used: sphere-spectroradiometer, sphere-photometer, 2π or 4π geometry, goniophotometer
- Designation and type of reference standard used: wattage; 2π or 4π total spectral radiant flux standard lamp or standard LEDs; CCT for white LED standard, or peak wavelength for single-color LED standard
- Correction factors applied (e.g., self-absorption, spectral mismatch)
- Optical measurement conditions (diameter of sphere, radius of goniophotometer)
- Measurement results: input current, V_F , power, total luminous flux, and/or total radiant flux
- Color quantities: chromaticity coordinates, CCT, D_{uv} , and/or CRI, as applicable for white DUTs; dominant wavelength, peak wavelength, or centroid wavelength, as applicable for colored LEDs
- Spectral power distribution (if applicable)
- Bandwidth of spectroradiometer, if spectral distribution and/or color quantities are reported
- Other equipment used
- Statement of uncertainties (if required)
- Deviation from standard operating procedures, if any

Table 10-1. Example Table of Measurement Results

Measurement Quantity	Result	Expanded Uncertainty ($k = 2$)
Method of measurement	Single-pulse mode	-
Pulse condition	Pulse width 20 ms	-
T_J	55 °C	-
Photometric method	Sphere spectroradiometer (2 π)	-
Reference standard	75 W total spectral radiant flux, standard lamp	-
Correction factors	Self-absorption	-
Photometric condition	0.5-m sphere	-
Input current	350.0 mA	0.5%
V_F	4.083 V	0.5%
Electrical power	1.429 W	0.7%
Total luminous flux	67.0 lm	4.2%
Luminous efficacy	46.9 lm/W	4.3%
Total radiant flux	0.2127 W	4.2%
Radiant efficiency	14.9%	4.3%
Chromaticity coordinate u'	0.2407	0.0014
Chromaticity coordinate v'	0.5040	0.0027
Correlated color temperature	3421 K	60 K
D_{uv}	-0.0066	0.0014
CRI, R_a	79.3	1.1

Annex A – Determination of Temperature Stability

To determine whether the DUT temperature is stable and ready for measurement, the DUT is driven at a low monitor current, I_M , ideally lower than 0.1 % of the DUT rated current and not more than 0.5% of the rated current. The DUT V_F is measured and plotted over the stabilization time. A stable condition is reached when the expected further V_F change before reaching the thermal equilibrium (estimated from the trend of plotted V_F stabilizing curve) is small enough so that the corresponding error in the measured optical quantity (for example total luminous flux) is acceptable. This measurement error can be calculated as follows:

- Divide the sum of the further V_F change by the

S_{VF} at the measurement current, to obtain the temperature error, ΔT . (Note: This S_{VF} can be an average or typical value for the part type.)

- Multiply the temperature coefficient of the optical quantity by the obtained ΔT to obtain the measurement error resulting from the ΔT .
- When this calculated error is acceptable, the DUT is considered stabilized.

When applicable, stability of a DUT temperature can also be determined directly by performing continuous optical measurements and plotting the measurement results after the DUT has been turned on. If the further change of measured optical quantity before reaching the thermal equilibrium (estimated from the trend of plotted stabilizing curve) is small compared to the allowed overall measurement uncertainty, the DUT temperature is considered stable.

ANNEX B – Determination of Transient Voltage Anomaly Effect

As shown in **Figure B-1**, the transient voltage anomaly effect distorts the LED's V_F during the first milliseconds of operation, superimposing a decaying voltage transient on top of the LED's normal voltage vs. time curve. As is well documented in JESD51-51,⁵ the V_F vs. t curve is normally a function only of the DUT's T_J . Thus, the presence of a transient voltage anomaly effect can be inferred if the V_F vs. t curve is determined not to correspond to T_J .

To determine whether a V_F vs. t curve does not correspond to T_J , V_F is plotted vs. square root of time, t , for two temperatures, e.g., 25 °C and 26 °C. The two curves may then be used to determine the calculated $\Delta T_{J,rise}$ based upon V_F . This may then be compared with

estimated $\Delta T_{J,rise}$ based upon radiant flux or irradiance. This is illustrated in the example in **Figure B-2**, which shows a plot of an LED's V_F vs. square root of time, t , curve at two case temperatures, 25 °C and 26 °C. Voltage is on the vertical axis and the square root of time, t , on the horizontal axis. Also plotted is the instantaneous S_{VF} ,

$$S_{VF} = \frac{V_{F, T_{TCM}=26\text{ °C}} - V_{F, T_{TCM}=25\text{ °C}}}{26 - 25} \quad (\text{B-1})$$

A number of conclusions can be drawn from this plot:

- The curves are approximately linear; this corresponds to a device with a square junction area receiving constant heating.
- The $\Delta T_{J,rise}$ calculated from the starting and ending points of the V_F curves and the S_{VF} , is about 7.5 °C.

The $\Delta T_{J,rise}$ of 7.5 °C is reasonable for this LED and its mount, when compared with calculations based upon radiant flux change or irradiance change. This DUT shows no obvious transient voltage anomaly effect.

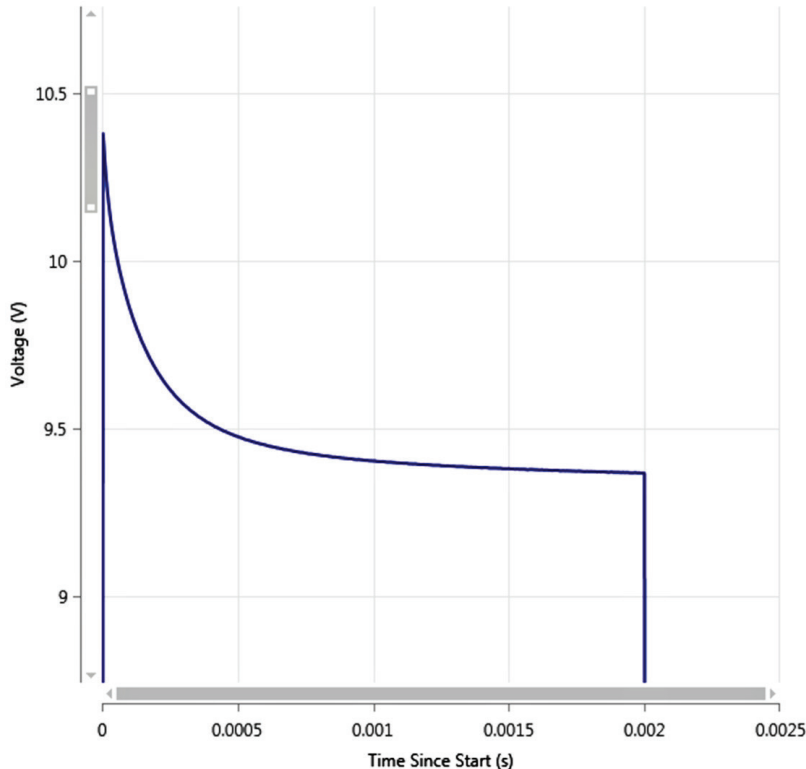


Figure B-1. LED V_F vs. time: 20-ms, 200-mA current pulse, showing 900-mV transient voltage anomaly effect.

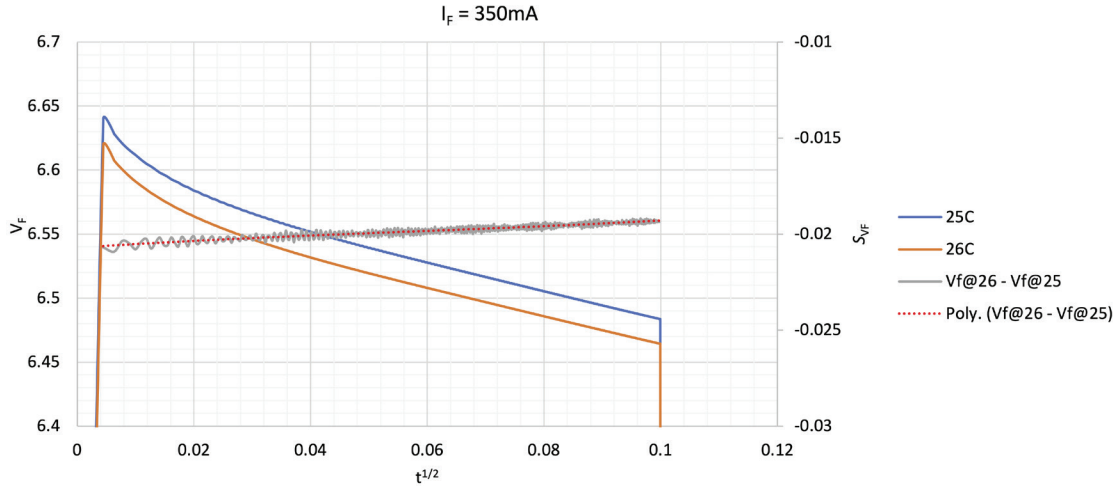


Figure B-2. Example of an LED showing no obvious transient voltage anomaly effect and a reasonable calculated $\Delta T_{J,rise}$ of 7.5 °C.

Figure B-3 shows a plot of a similar LED's V_F vs. square root of time, t , curve again at two case temperatures of 25 °C and 26 °C. This plot shows:

- The curves are not linear; this could mean a more complex thermal path, or it could be the transient voltage anomaly effect.
- The $\Delta T_{J,rise}$ calculated from the starting and ending points of the V_F curves and the S_{VF} , is about 75 °C.

Because the possible existence of the transient voltage anomaly effect, the measured forward voltage curve, $V_F(t)$, should be compared to the relative radiant flux curve, $\Phi_{e,rel}(t)$, or the irradiance curve, $E_{e,rel}(t)$, measured using a radiometer with flat spectral responsivity.⁶ The deviation of $V_F(t)$ from $\Phi_{e,rel}(t)$ or $E_{e,rel}(t)$ indicates existence of the transient voltage anomaly effect; therefore, the radiant flux should be used to infer T_J .

The $\Delta T_{J,rise}$ of 75 °C is not reasonable for this LED and its mount. Thus, this DUT has an obvious transient voltage anomaly effect.

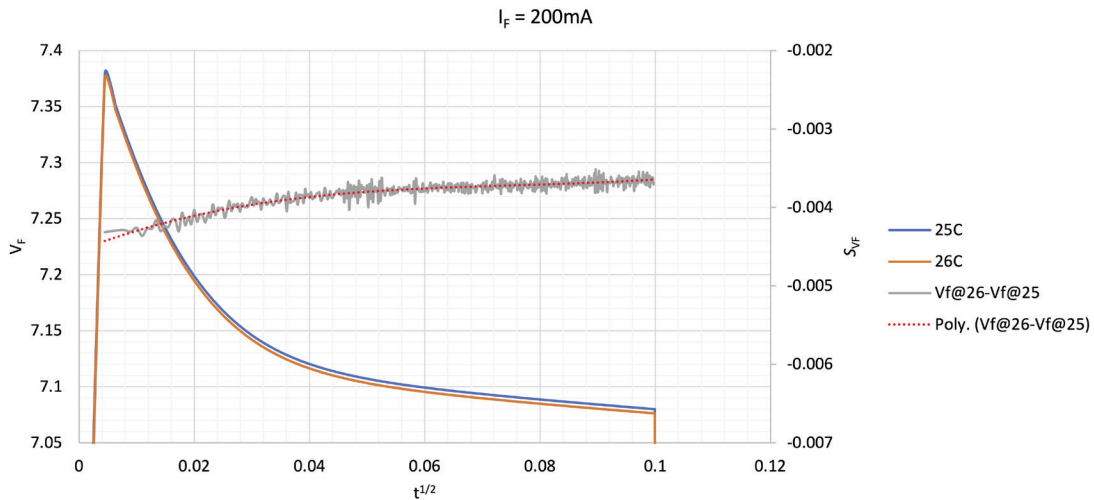


Figure B-3. An example of an LED with a transient voltage anomaly effect and an unreasonable calculated $\Delta T_{J,rise}$ of 75 °C.

ANNEX C – Assessing T_j Rise in Continuous-Pulse Mode Using Transient Thermal Resistance

In the case of CP operation, the maximum rise of the T_j , $\Delta T_{j, \text{rise}, \text{max}}$, immediately before the fall of the pulse can be calculated by 7:

$$\Delta T_{j, \text{rise}, \text{max}} = P \cdot \left\{ \left[\frac{t_{pw}}{t_{pp}} \cdot R_{th} \right] + \left[\left(1 - \frac{t_{pw}}{t_{pp}} \right) \cdot r_{th}(t_{pw} + t_{pp}) \right] + r_{th}(t_{pw}) - r_{th}(t_{pp}) \right\}, \quad (\text{C-1})$$

where:

t_{pw} = the pulse width, in seconds

t_{pp} = the pulse period, in seconds

R_{th} = the thermal resistance of the LED (at the ambient air temperature), in $^{\circ}\text{C}/\text{W}$

$r_{th}(t)$ = the transient thermal resistance function of pulse width, in $^{\circ}\text{C}/\text{W}$

P = the electrical power of the LED, in watts

An example of $r_{th}(t)$ is shown in **Figure C1** and calculated $\Delta T_{j, \text{rise}, \text{max}}$ is shown in **Figure C2**.

For example, for a 3-watt LED with a pulse width $t_w = 0.05$ ms and a pulse period $t_{pp} = 5$ ms (i.e., 200 Hz with duty cycle 1/100), and using the $r_{th}(t)$ data in **Figure C-1**, the maximum rise of T_j will be $\Delta T_{j, \text{rise}, \text{max}} \approx 2.3$ $^{\circ}\text{C}$ as shown in **Figure C-2**. The error in measured photometric and colorimetric values can be estimated from this $\Delta T_{j, \text{rise}, \text{max}}$ value and the data showing temperature dependence of the photometric quantities, thereby allowing corrections to be made or the uncertainty contribution to be calculated.

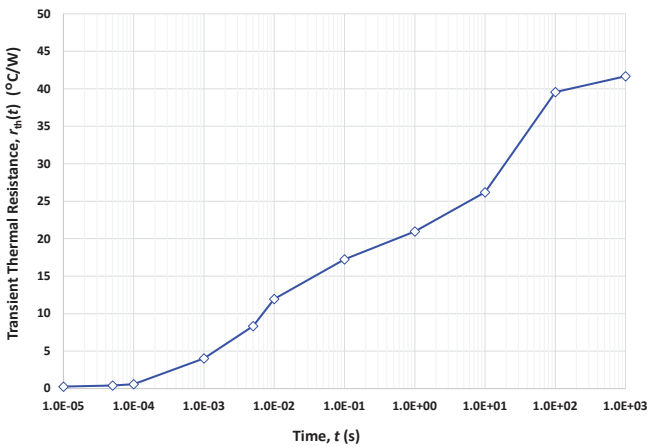


Figure C-1. The transient thermal resistance of a typical LED that is not mounted on a heat sink.

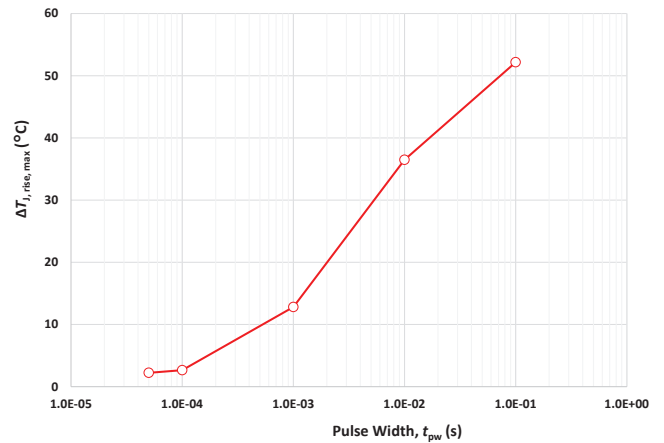


Figure C-2. The calculated maximum rise of T_j , $\Delta T_{j, \text{rise}, \text{max}}$, with 1 % duty cycle.

Annex D – Assessing T_J Rise During Pulsed Operation Using a DUT Temperature Sensitive Parameter

D.1 Introduction

Assessing the magnitude of T_J rise, $\Delta T_{J, \text{rise}}$, during a pulsed operation is useful for determining the error associated with a measurement (see **Equation 8-2 in Section 8.1.1 SP-1 Method**) and for improving the measurement—for example, by adjusting the TCM temperature to a lower setpoint to compensate for the difference between the actual T_J and the specified T_J .

Most LED and laser parameters exhibit temperature dependence. For example, for LEDs and lasers, V_F decreases with temperature, as does optical output. For some LED types, such as amber-colored aluminum gallium indium phosphide (AlGaInP) LEDs, the temperature dependence of optical output is quite high, approximately $-1.5\%/^{\circ}\text{C}$ for a typical device. For these highly temperature sensitive devices, it is vital to understand the actual T_J during measurements. But even less temperature-sensitive devices may exhibit large errors due to T_J rise during pulsed measurements. Pre-2017 LEDs often incorporated a large copper plate known as a “heat slug” to spread heat to the attachment. When these LEDs were tested in an unmounted state with 20-ms long pulses, only modest temperature rises occurred due to the thermal capacity of the slug. But post-2017 LEDs often feature optimized packages that omit or remove this slug. With no external sink to dissipate the heat, their T_J rises dramatically during a long-pulse measurement.

D.2 Pulsed T_J : Two Components

The T_J during a pulsed measurement is influenced by two factors: 1) the initial T_J prior to the pulse; and 2) the transient heating of the junction during the pulse. Both these components should be considered.

D.2.1 Initial T_J Prior to the Pulse. If the test method involves only a single pulse, then the initial T_J prior to the pulse is determined only by the DUT's immediate thermal environment. If a precision TCM is used, and it is operating at a setpoint close to

room temperature (e.g., 25°C), the initial T_J should be very close to the TCM setpoint. However, at elevated temperature setpoints, heat loss to the surrounding environment may result in a significant negative T_J error. To measure this error, the device should be allowed to come to thermal equilibrium with no power, and then the temperature measurement point (TMP) may be measured with a thermocouple or other sensor. The measured temperature, rather than the setpoint temperature, should then be used as the initial T_J prior to the pulse.

For CP, DCP, and M-DCP devices that use multiple pulses, initial T_J is also influenced by average heating, even at the low (1% to 2%) duty cycles that are typically used. For these measurement methods the initial T_J can be measured using a variation of the JEDEC electrical test method (ETM) described in **Section D.3 ETM Method 1**.

D.2.2 Transient T_J During the Pulse. Transient heating during the pulse can also be measured using the JEDEC ETM. However, if a DUT exhibits the transient voltage effect, then the heating shall be evaluated using an optical measurement parameter that is temperature sensitive, such as radiant flux rather than the V_F . The following sections utilize V_F , but the same calculations could be applied to an optical measurement parameter.

D.3 ETM Method 1: Transient T_J Measurement Using Separate Heating and Measurement Currents

The classic electrical test method (ETM) uses a two-step process to infer junction temperature, T_J . First the DUT's V_F as a function of temperature is characterized using a low DC “measurement” current that biases the diode junction on but does not heat the device significantly. Next, the DUT is operated with a combination of the pulsed “heating” current plus the DC measurement current. This can be accomplished with a pulsed-current source or SMU that is equipped with a secondary bias current source. Finally, the DUT V_F is digitized, and the V_F just before and after the pulse is compared with the DUT's characterized V_F vs. temperature function to infer T_J .

Figure D-1 shows a typical transient T_J ETM V_F capture. The gray dashed horizontal line is the V_F associated with DC operation at 26°C and at the DC-measurement

current level. The orange dashed line is the V_F with DC operation at 25 °C. The value of the difference between these two voltages equals the DUT's temperature coefficient S_{VF} . It is important to note that 25 °C and 26 °C were chosen because they are close to the nominal test T_J of 25 °C. For this DUT, a 3-W blue LED, the S_{VF} is:

$$S_{VF} = \frac{2.79520 - 2.79787}{26 - 25} = -0.00267 \quad (\text{D-1})$$

Using the DC V_F value for 25 °C and the calculated S_{VF} , the 10- μ s pre-pulse T_J is:

$$T_{J,\text{pre-pulse}} = 25 + \frac{V_{F,\text{pre-pulse}} - V_{F,25^\circ\text{C}}}{S_{VF}} = 25 + \frac{2.79670 - 2.79787}{-0.00267} = 25.44 \quad (\text{D-2})$$

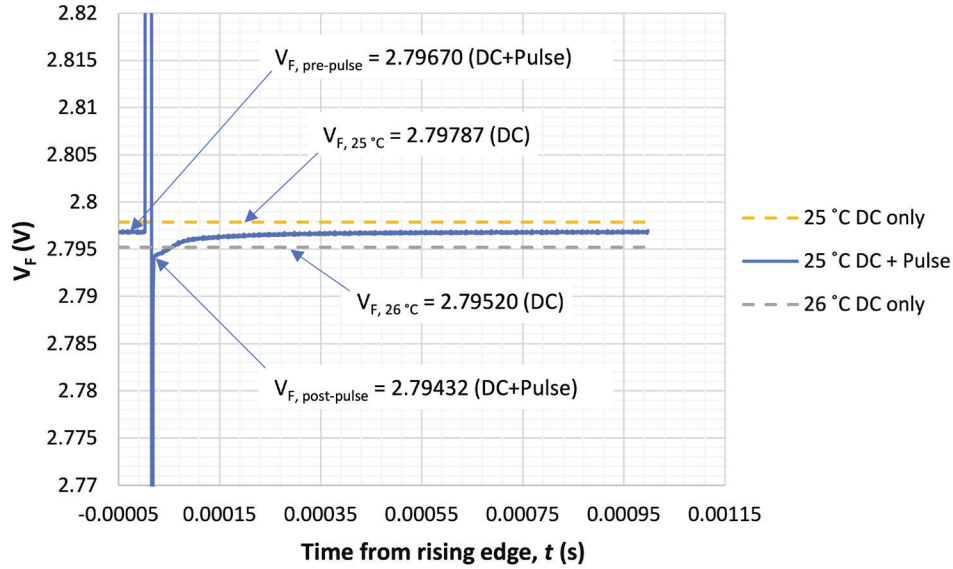


Figure D-1. ETM Method 1, determination of average transient $T_{J,\text{avg}}$, given a 10- μ s pulse and 1% duty cycle.

The post-pulse T_J can similarly be calculated using the post-pulse V_F . In this case, the first valid voltage sample after the electrical transient should be used. While the JEDEC ETM method allows extrapolation using a square-root-of-time horizontal axis, this practice can yield measurements that miss some initial heating. For best accuracy, the electrical transient should be minimized so that samples very close to the falling edge of the pulse can be used; alternatively, an optical parameter should be used to infer temperature.

10- μ s post-pulse T_J :

$$T_{J,\text{post-pulse}} = 26 + \frac{V_{F,\text{post-pulse}} - V_{F,26^\circ\text{C}}}{S_{VF}} = 26 + \frac{2.79432 - 2.79520}{-0.00267} = 26.33 \quad (\text{D-3})$$

Note: As it is closer to the final T_J , the 26°C DC V_F is used instead of the 25°C DC V_F .

To a first approximation, the average transient junction temperature, $T_{J,\text{avg}}$, during the pulse is the average of the pre-pulse T_J and the post-pulse T_J :

$$T_{J,\text{avg}} = \frac{T_{J,\text{pre-pulse}} + T_{J,\text{post-pulse}}}{2} = \frac{25.44 + 26.33}{2} = 25.89 \quad (\text{D-4})$$

This approximation ignores the fact that LEDs and lasers typically heat faster during the early part of a pulse and more slowly during the latter part of the pulse. Because of this, the actual average transient $T_{J,\text{avg}}$ is greater than the average of the pre-pulse T_J and the post-pulse T_J .

This classic ETM may be used to assess heating with longer or shorter pulses, as shown in **Figures D-2** through **D-4** and **Table D-1**.

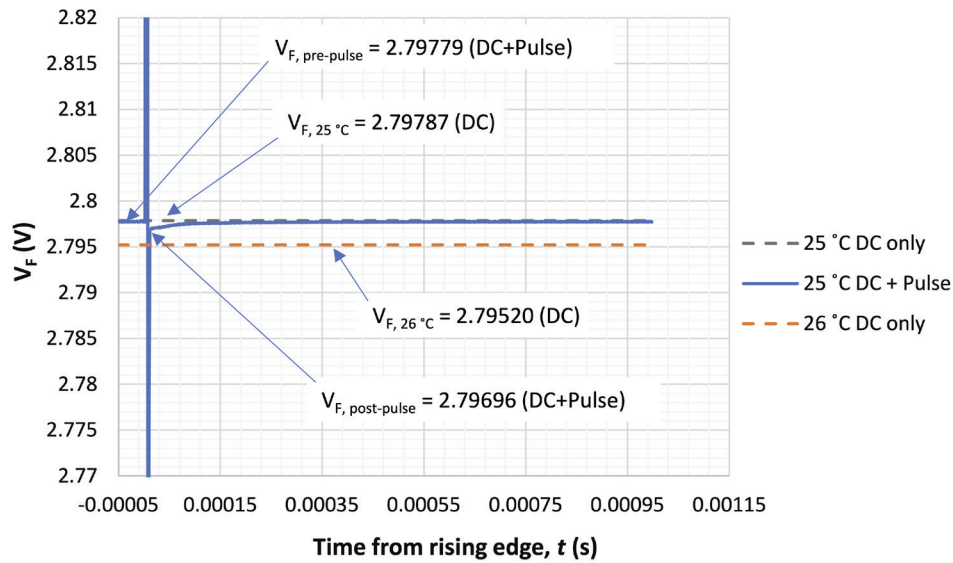


Figure D-2. ETM Method 1, determination of average transient $T_{j,avg}$ given a 2.5- μ s pulse and 1% duty cycle.

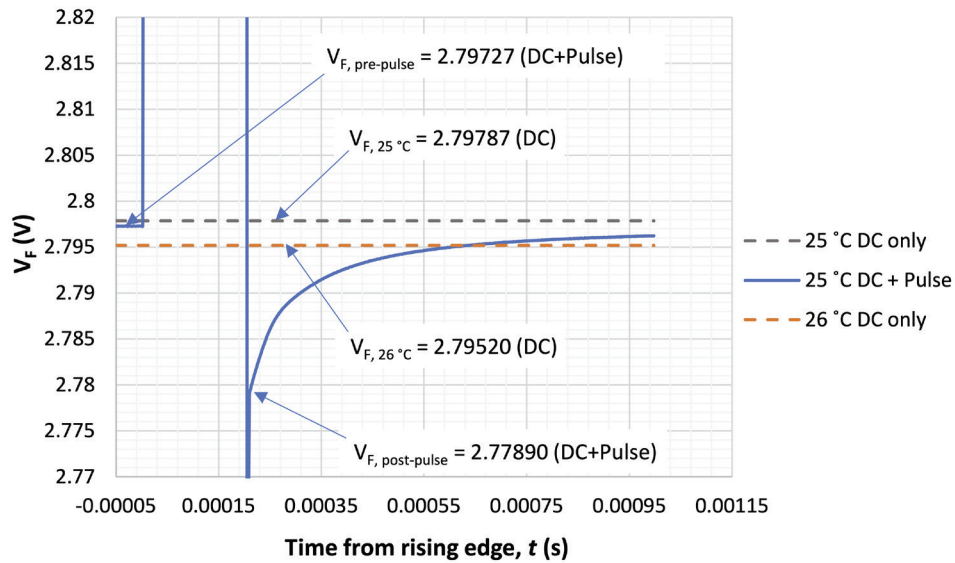


Figure D-3. ETM Method 1, determination of average transient $T_{j,avg}$ given a 200- μ s pulse and 1% duty cycle.

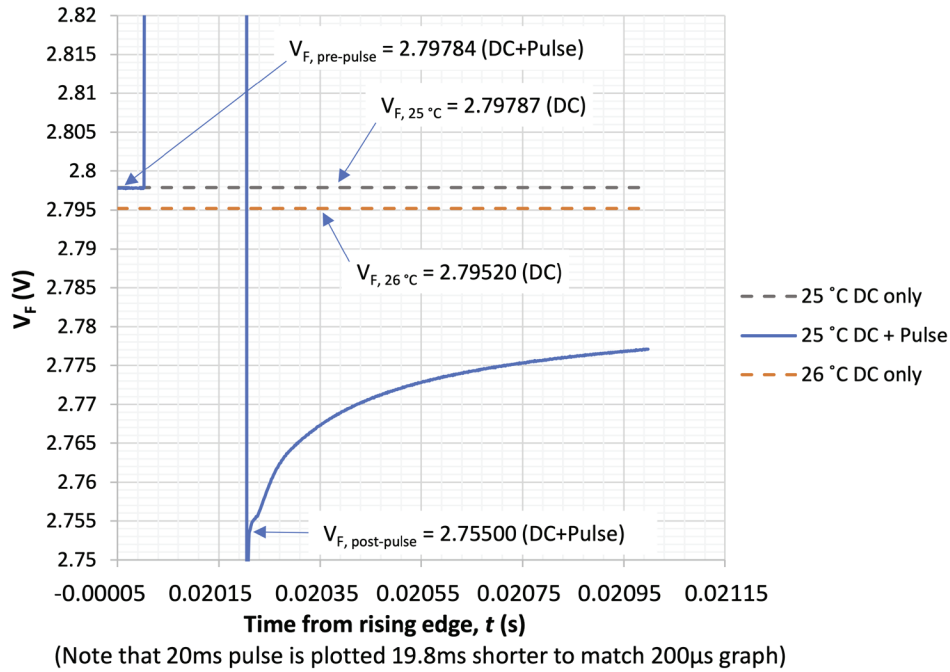


Figure D-4. ETM Method 1, determination of average transient $T_{J,avg}$, given a 20-ms pulse and 1% duty cycle.

Table D-1 summarizes the calculation of the average transient $T_{J,avg}$ for the various pulse widths.

Table D-1. Blue LED Average Transient $T_{J,avg}$ for Various Pulse Widths, 1% Duty Cycle. Set $T_J = 25\text{ }^{\circ}\text{C}$.

Pulse Width (μs)	$V_{F, 25\text{ }^{\circ}\text{C DC}}$ (V)	$V_{F, 26\text{ }^{\circ}\text{C DC}}$ (V)	S_{VF} (V/ $^{\circ}\text{C}$)	$V_{F, \text{pre-pulse}}$ (V)	$V_{F, \text{post-pulse}}$ (V)	$T_{J, \text{pre-pulse}}$ ($^{\circ}\text{C}$)	$T_{J, \text{post-pulse}}$ ($^{\circ}\text{C}$)	$\Delta T_{J, \text{rise during pulse}}$ ($^{\circ}\text{C}$)	$T_{J, \text{avg}}$ ($^{\circ}\text{C}$)
2.5	2.79787	2.79520	-0.00267	2.79779	2.79696	25.03	25.34	0.31	25.2
10	2.79787	2.79520	-0.00267	2.79670	2.79432	25.44	26.33	0.89	25.9
200	2.79787	2.79520	-0.00267	2.79727	2.77890	25.22	32.10	6.88	28.7
20,000	2.79787	2.79520	-0.00267	2.79784	2.75500	25.01	41.06	16.0	33.0

The relative measurement error for flux, $\epsilon_{\phi, \text{rel}}$ (%), associated with the junction heating can be calculated using **Equation 8-2**. For this blue LED, the luminous flux sensitivity coefficient, $c_{T_J-\phi}$, is $-0.17\text{ }(\%/^{\circ}\text{C})$. Using the $T_{J,avg}$ values from **Table D-1**, the errors due to junction heating during the pulse are shown in **Table D-2**.

Table D-2. Blue LED Luminous Flux Error Due to Junction Heating for Various Pulse Widths

Pulse Width (μs)	$T_{J, \text{nominal}}$ ($^{\circ}\text{C}$)	$T_{J, \text{avg}}$ ($^{\circ}\text{C}$)	T_J error ($^{\circ}\text{C}$)	$c_{T_J-\phi}$ ($\%/^{\circ}\text{C}$)	$\epsilon_{\phi, \text{rel}}$ (%)
2.5	25	25.2	0.19	-0.17	-0.03
10	25	25.9	0.88	-0.17	-0.15
200	25	28.7	3.66	-0.17	-0.62
20,000	25	33.0	8.03	-0.17	-1.37

D.4 ETM Method 2: Transient T_J Measurement Using Only Heating Current

ETM Method 2 uses only the V_F sampled during the pulsed heating current to infer temperature. This eliminates the need for a secondary bias current source, and it allows the instantaneous T_J rise throughout the pulse to be calculated. It does not allow the pre-pulse T_J to be determined during the pulse on time.

In this method, captures of the DUT V_F at two different temperatures, T_1 and T_2 , are digitized. Only the pulse on time is captured. The SVF is determined from the average difference in V_F of the two captures, or from a common point in both captures (such as the minimum V_F point before the falling edge). It is important to note that this S_{VF} is often significantly different from the measurement-current S_{VF} (See **Equation D-1**), as it is for a much higher current level. The maximum $T_{J,\text{rise}}$, $\Delta T_{J,\text{rise,max}}$, during the pulse can then be calculated using the maximum and minimum V_F values taken from one of the captures:

$$\Delta T_{J,\text{rise,max}} = \frac{V_{F,\text{min}} - V_{F,\text{max}}}{S_{VF}} \quad (\text{D-5})$$

$$S_{VF} = \frac{V_{F,T_2\text{min}} - V_{F,T_1\text{min}}}{T_2 - T_1} \quad (\text{D-6})$$

The instantaneous T_J rise for any measurement point i ($i = 1, n$) may also be calculated as:

$$\Delta T_{J,\text{rise},i} = \frac{V_{F,i} - V_{F,0}}{S_{VF}} \quad (\text{D-7})$$

The average T_J rise, $\Delta T_{J,\text{rise,avg}}$, is then the average of all the measurement points during the pulse:

$$\Delta T_{J,\text{rise,avg}} = \frac{\sum_{i=1}^n \Delta T_{J,\text{rise},i}}{n} \quad (\text{D-8})$$

And the average transient junction temperature, $T_{J,\text{avg}}$, is:

$$T_{J,\text{avg}} = T_{J,0} + \frac{\sum_{i=1}^n \Delta T_{J,\text{rise},i}}{n}, \quad (\text{D-9})$$

where $T_{J,0}$ is the junction temperature just before the pulse.

It is important to note that the $T_{J,\text{avg}}$ obtained using **Equation D-9** is the true average transient junction temperature. The calculated average T_J rise from the T_J set point, $\Delta T_{J,\text{rise,avg}}$, is typically 30% to 60% greater than the endpoints-only average $\Delta T_{J,\text{rise,avg}}$ obtained using the ETM Method 1. Method 2 does not determine the initial T_J prior to the pulse. If average heating is significant, then ETM Method 1 should be used to measure the T_J rise prior to the pulse. This temperature rise should be added to the average T_J rise measured in Method 2.

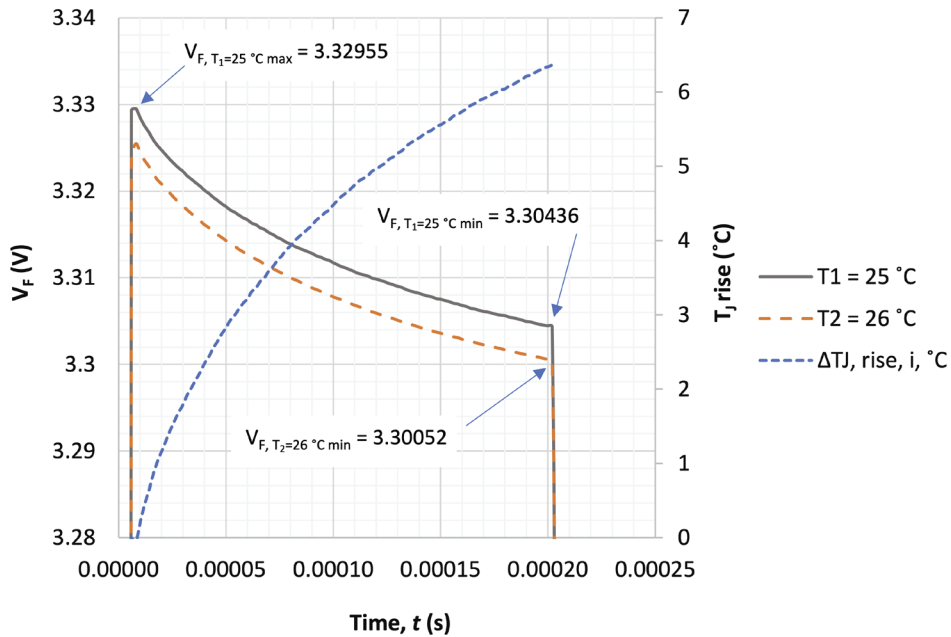


Figure D-5. ETM Method 2: V_F captures, with calculated instantaneous T_J rise.

Figure D-5 illustrates ETM Method 2. The gray curve shows the capture at 25 °C, and the dashed orange curve shows the capture at 26 °C. The dashed blue curve represents the calculated instantaneous T_J rise, $\Delta T_{J, rise, i}$. An example of the measurement result using this ETM Method 2 is shown in **Table D-3**.

Table D-3. 200- μ s Blue LED Average Transient $T_{J, avg}$, Obtained With ETM Method 2. Set $T_J = 25$ °C.

Pulse Width (μ s)	$V_F, T_1 = 25$ °C min (V)	$V_F, T_2 = 26$ °C min (V)	S_{VF} (V/°C)	$\Delta T_{J, rise, max}$ during pulse (°C)	$\Delta T_{J, rise, avg}$ during pulse (°C)	$T_{J, avg}$ during pulse (°C)
200	3.30436	3.30050	-0.00386	6.5	4.2	29.2

INFORMATIVE REFERENCES

- 1 Illuminating Engineering Society. ANSI/IES TM-30-20, Technical Memorandum: IES Method for Evaluating Light Source Color Rendition. New York: IES; 2020.
- 2 International Commission on Illumination (CIE). CIE 225:2017, Optical Measurement of High-Power LEDs. Vienna: CIE; 2017.
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- 5 JEDEC Solid State Technology Association. JEDEC JESD 51-51, Implementation of the Electrical Test Method for the Measurement of Real Thermal Resistance and Impedance of Light-Emitting Diodes With Exposed Cooling. Arlington, Virg.: JEDEC; 2012.
- 6 Thorlabs. Response-Flattening Filters for Silicon Photodiodes. Online: https://www.thorlabs.com/newgrouppage9.cfm?objectgroup_id=11424. (Accessed 2023 Oct 5).
- 7 Toshiba Electronics Devices & Storage Corporation. Bipolar Transistors Application Note: Bipolar transistors thermal stability and thermal design. Tokyo: Toshiba; 2021.

Process for Change to an ANSI/IES Standard under Continuous Maintenance

This standard is maintained under continuous maintenance procedures, for which IES has an established and documented program for regular publication of addenda or revisions, including procedures for timely, documented, consensus action on requests for change to any part of the standard. Committee consideration will be given to proposed changes by June 30 of any given year for proposed changes received by the IES Director of Standards no later than December 31 of the previous year.

Submittal Format

Proposed changes must be submitted to the IES Director of Standards in the announced published format. However, changes may be accepted in an earlier published format, if the differences are immaterial to the proposed change submittal. If the Director of Standards concludes that a current form must be utilized, the proposer may be given up to 20 additional days to resubmit the proposed changes in the current format.

Specific changes in the text or values are required and must be substantiated. Any change proposals that do not meet these requirements will be returned to the proposer. Supplemental background documents to support changes submitted may be included.

Submission to the Committee Chair

The Director of Standards shall forward proposed changes received on appropriate forms to the committee chair for assigning to committee members (responders) to develop responses to submitters of proposed changes.

Review and Clarification

Responders shall review proposals and should contact the proposer if necessary for clarification.

Response Recommendation

Designated responders shall draft a recommended committee response, including any recommended changes to the standard. The 'responders' recommended responses shall be submitted to the committee chair in electronic form usable by Society Staff, including any recommended change to the standard in response to proposals received.

Options for Committee response are limited to:

- a) Proposed change accepted for public review without modification
- b) Proposed change accepted for public review with modification
- c) Proposed change accepted for further study
- d) Proposed change rejected

The responders shall provide reasons for any recommendation other than option (a) above.

The designated responders shall not recommend option (c) unless the further study can be completed by October 1 of that year, and providing the Committee can then vote for option (a), (b), or (d) no later than November 15 of that year.

Editing

The Committee chair or his or her designee shall edit the draft responses and circulate the edited drafts to the committee for review.

Form for Proposing Change to an ANSI/IES Standard under Continuous Maintenance

NOTE:

Use a separate form for each comment. Submit to the Director of Standards, IES, 85 Broad St, FL 17, New York NY 10004.
Email: standards@ies.org. Fax: 212-248-5017.

1. Submitter: _____
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I hereby grant the Illuminating Engineering Society (IES) the non-exclusive royalty rights, including non-exclusive rights in copyright, in my proposals. I understand that I acquire no rights in publication of the standard in which my proposals in this, or other analogous, form are used. I hereby attest that I have the authority and am empowered to grant this copyright release.

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2. Title of publications and year published _____
3. Clause (section), sub-clause or paragraph number; and page number: _____
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- ☐ Change to read as follows
 - ☐ Delete and substitute as follows
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Use underscore to show material to be added (added) and strikethrough for material to be deleted (~~deleted~~). Use additional pages if needed.

5. Proposed change:

6. Reason and substantiation:

Select as applicable:

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- ☐ Attachments or referenced materials cited in this proposal accompany this proposed change.

Please verify that all attachments and references are relevant, current, and clearly labeled to avoid processing and review delays. Please list your attachments here:



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